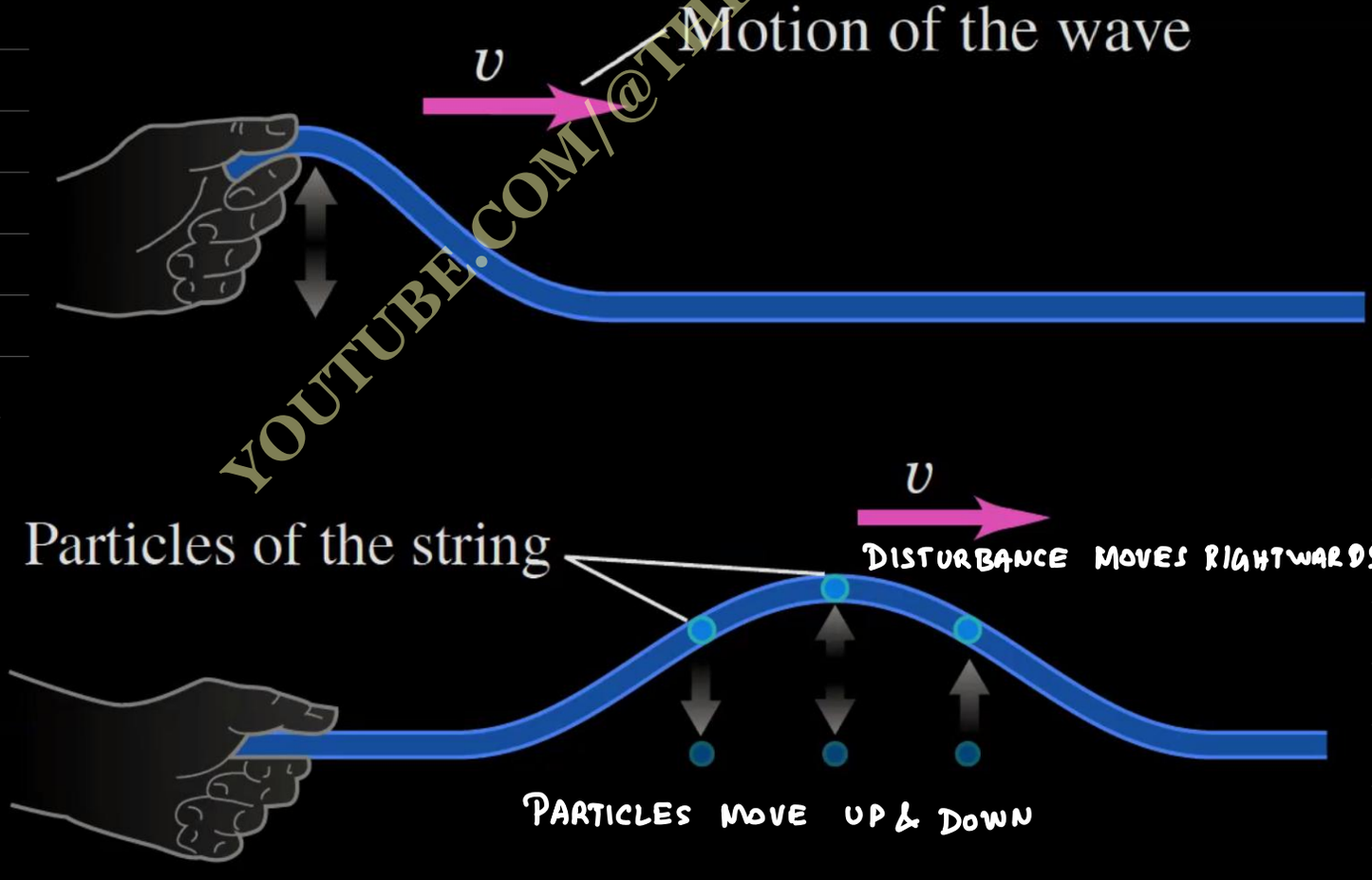
Mechanical Wave: A mechanical wave is a disturbance (particles (configuration) that travels through some material (known as medium) for the wave. A mechanical wave involves particles moving from their configuration and then getting back to their original configuration without bulk transport of materials. We shall particularly focus on one dimensional waves in which the disturbance travels only in a single direction such as waves on a string or a straight column of air though waves can also travel in 2D (e.g. waves on surface of a drum membrane, or sound waves emanating from a loud speaker (3D). Waves can be further be classified in to transverse, longitudinal or a combination (elliptical waves).

TRANSVERSE WAVES: When the direction of oscillations (or movements) of medium particles is perpendicular to the direction of propagation of disturbance, we call it transverse waves.



TRANSVERSE WAVE IN STRING

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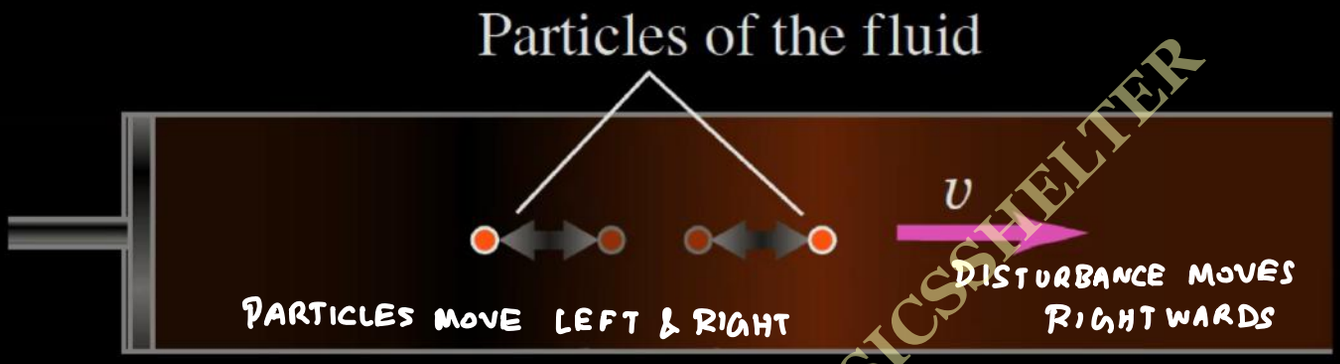
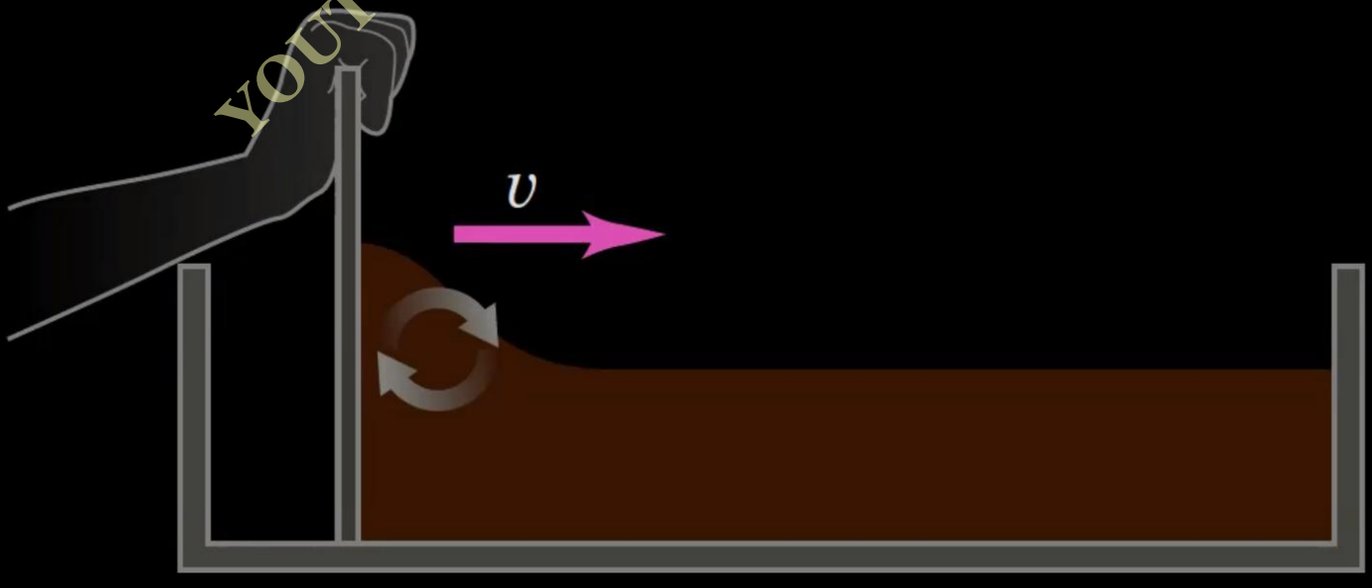
LONGITUDINAL WAVE: When direction of oscillations (or motion of particles) is along the direction of propagation of disturbance we call it longitudinal waves.



LONGITUDINAL WAVE IN AN AIR COLUMN.

ELLIPTICAL WAVES: If the particles of medium have components of motion along the propagation of disturbance as well as perpendicular to it with same frequency we have elliptical waves (can also have lissajous waves if frequencies are different in different directions).



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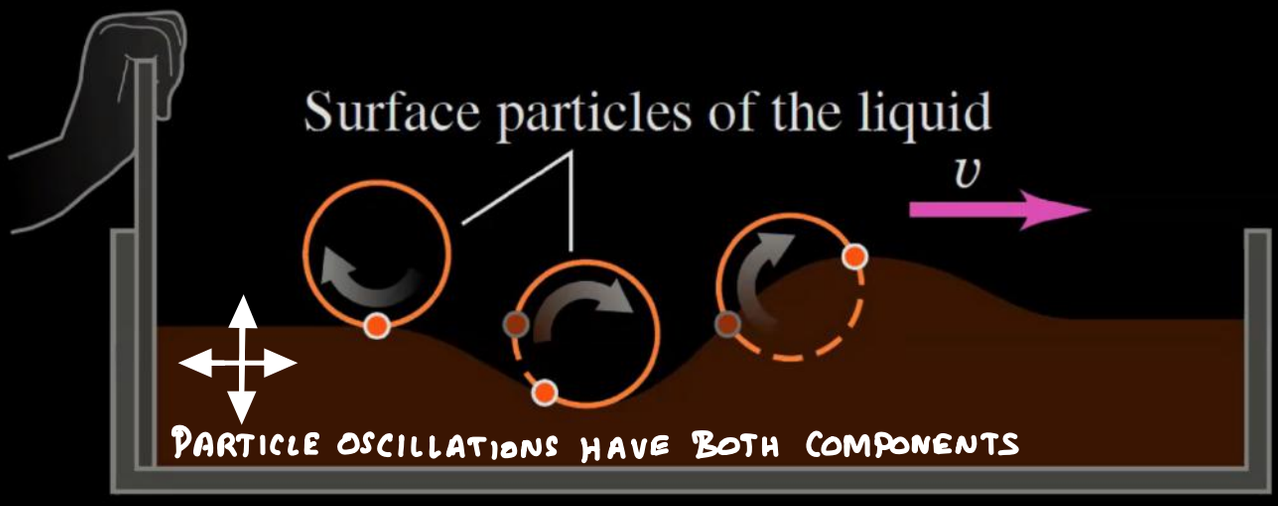
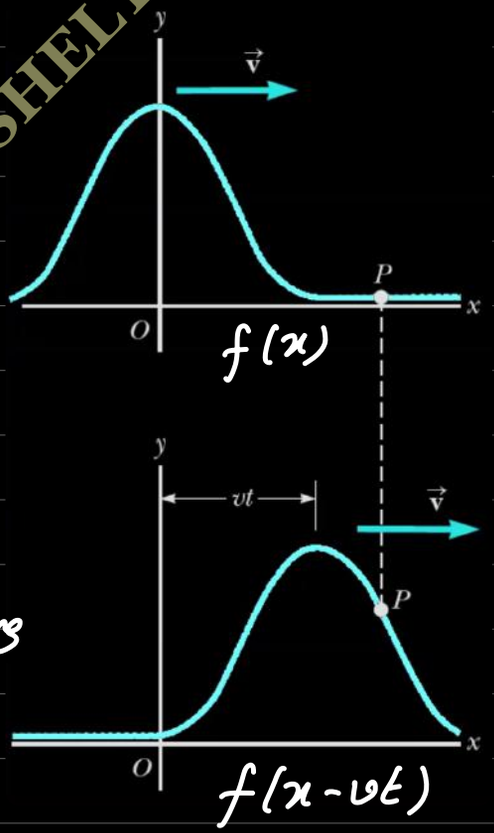


FUNCTIONAL FORM DESCRIBING A TRAVELLING WAVE:

Recall from coordinate geometry that functions $f(x)$ and $f(x-x_0)$ have exactly same shape except that in the latter case the curve is shifted to right by an amount x_0. Now if we put x_0 a function of time $x_0 = vt$ where v is a constant then we get a curve whose shape changes with time (shift a distance vt in a time t). This is the form of function which can represent a disturbance travelling in a string. Thus any function of the form $y = f(x-vt)$ represents a wave travelling to the right with a velocity v. Similarly a function of the form $f(x+vt)$ represents a wave travelling to the left with speed v. Here:



x: x coordinate of the string particle.

y: y coordinate or displacement of the particle located at x.

t: time

v: speed of the wave.

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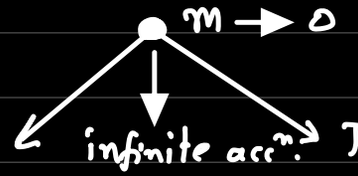
RESTRICTIONS ON $f(x-vt)$ or $f(x+vt)$: To represent an actual physical wave, the function f must satisfy following properties:

① f should be continuous (otherwise string will break)

② f should be differentiable (∵ sudden slope change would cause infinite accelerations).



③ $f(x,t)$ must be bounded (Any material particle cannot have infinite displacement).

EXAMPLE: For the following functions find the wave velocity if it can represent physical wave:

(a) $y = e^{-(4x-3t)^2}$ (b) $y = e^{(x-4t)}$

(c) $y = A\sin(kx+\omega t)$ (d) $y = \dfrac{1}{(x-5t)}$

(e) $y = \dfrac{1}{(3x-4t)^2+5}$

Solution: (a) $e^{-(4x-3t)^2}$ is bounded, continuous & differentiable it can be written as

$$y = e^{-16(x-3t/4)^2} = f(x-vt)$$

By comparison, $v = +3/4$ ⇒ moves to the right at 3/4

(b) $y = e^{(x-4t)}$ is not bounded ∴ does not represent a physical wave. (Mathematical velocity is +4).

(c) $y = A\sin(kx+\omega t)$ it is continuous bounded and differentiable ∴ can represent a physical wave. $y = A\sin k\left(x+\frac{\omega}{k}t\right)$ comparing with $y = f(x-vt)$, $v = -\frac{\omega}{k}$, thus wave travels leftwards at $v = \omega/k$.

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(d) $y = \frac{1}{x-5t}$ is not bounded $\therefore$ can't represent a physical wave (goes to ∞ at $x = 5t$). Mathematical speed is $+5$.

(e) $y = \frac{1}{(3x-4t)^2+5}$ is bounded, continuous & differentiable

$\therefore$ can represent a physical wave. $y = \frac{1}{9(x-\frac{4}{3}t)^2+5}$

$= f(x-vt)$

$\therefore \quad v = +4/3$ Ans.

FINDING WAVE EQUATION FROM SNAPSHOT AT $t = t_0$: If $f(x)$ is the snapshot of a wave travelling to the right at a speed v, then the shape of the wave will be shifted to the right by an amount $v(t-t_0)$ at a general time (from the shape at $t = t_0$) thus at a general time, shape will be $f(x - v(t-t_0))$. Similarly if the wave were travelling to the left the shape will be $f(x + v(t-t_0))$. In general combining two forms:

$$\boxed{y = f(x \pm v(t-t_0))}$$

EXAMPLE: Shape of two waves at $t = 2$ sec is given as

(a) $y = e^{-x^2}$ and (b) $y = 1/(1+x^2)$

find the wave equations if first wave is travelling to the left at $5\,ms^{-1}$ and the second wave is travelling to the right a $5\,ms^{-1}$ (x & y are in metres).

Solution: (a) $f(x) = e^{-x^2} \Rightarrow f(x - v(t-t_0)) = e^{-[x+5(t-2)]^2}$. ($\because v = -5$) Ans

(b) $f(x) = 1/1+x^2 \Rightarrow f(x - v(t-t_0)) = 1/(1+[x-5(t-2)]^2)$

$\Rightarrow y = \frac{1}{1+(x-5t+10)^2}$ Ans.

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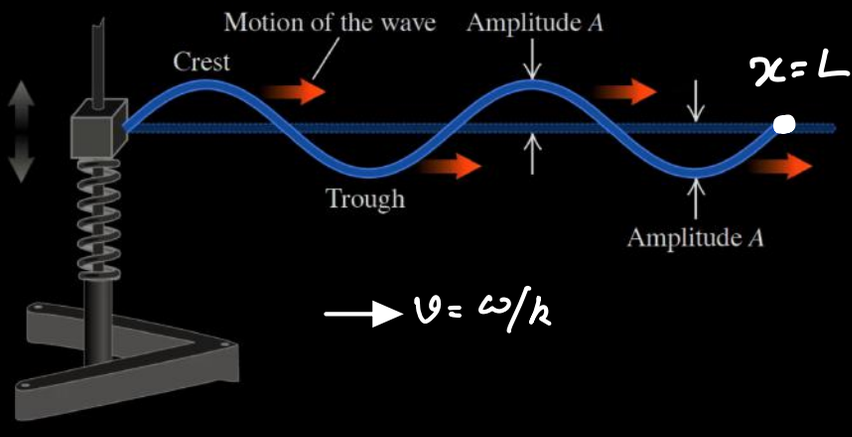
PAST AND FUTURE SHIFTING OF DISTURBANCES: Just as $f(x-x_0)$ represents right shifting of $f(x)$ by an amount x_0 similarly $f(t-t_0)$ for a time function represents shifting of $f(t)$ to a future time by a delay of t_0. i.e the behaviour of $f(t)$ is same as behaviour of $f(t-t_0)$ but at a time $t+t_0$. Let $f(t)$ be the disturbance of a point located at $x=x_0$ at a general time and let wave be travelling to right with a speed v, then for a particle located at a general x, the same disturbance will reach with a delay of $(x-x_0)/v$. Thus equation of wave in terms of general x and t can be written as:

$$y = f\left(t - (x-x_0)/v\right)$$ Similarly if the disturbance were travelling to the left, the wave equation could be written as:

$$y = f\left(t + (x-x_0)/v\right)$$, Combining the two forms together,

$$\boxed{y = f\left(t \mp (x-x_0)/v\right)}$$

EXAMPLE: Incident disturbance at $x=0$ is given as $y = A\sin(\omega t)$ it gets reflected at a boundary $x=L$ and undergoes a π shift upon reflection, without any change in amplitude. The reflected disturbance acts a a source for a reflected wave, Calculate:



(a) Incident wave equation.
(b) Incident disturbance at $x=L$
(c) Reflected disturbance at $x=L$
(d) Reflected wave equation.

Sol: $y_i(x=0) = A\sin(\omega t)$.

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(a) Applying future shift, $y_i(x,t) = f(t - x/v)$

$$\Rightarrow \quad y_i = A\sin\left(\omega\left(t - \frac{x}{(\omega/k)}\right)\right) = A\sin(\omega t - kx)$$

(b) @ $x = L$, $y_i = A\sin(\omega t - kL)$

(c) According to question, at $x = L$, y_r is π shifted or inverted (negation) of y_i $\Rightarrow$ $y_r = A\sin(\omega t - kL + \pi)$

$\Rightarrow y_r = A\sin(kL - \omega t)$. (@ $x = L$).

(d) Now to reach a general x, the reflected wave will take a time $(L-x)/v$. Now applying future shift:

$$y_r = A\sin\left(kL - \omega\left(t - \frac{(L-x)k}{\omega}\right)\right)$$

$$\Rightarrow y_r = A\sin(2kL - \omega t - kx) = A\sin(kx + \omega t - 2kL + \pi)$$

EXAMPLE: Repeat the previous problem if incident disturbance at $x = x_0$ is $f(t)$, reflecting boundary is still at $x = L$. v is known.

Sol: $y_i(t, x_0) = f(t)$, Applying future shift:

(a) $y_i(t, x) = f(t - (x - x_0)/v)$

(b) $y_i(t, L) = f(t - (L - x_0)/v)$

(c) $y_r = -y_i = -f(t - (L - x_0)/v)$,

Now applying future shift appropriately:

(d) $$y_r(x,t) = -f\left(t - \left(\frac{L-x}{v}\right) - \left(\frac{L-x_0}{v}\right)\right)$$

$$\Rightarrow y_r = -f\left(t + \frac{x}{v} - \frac{2L}{v} + \frac{x_0}{v}\right) \text{ Ans.}$$

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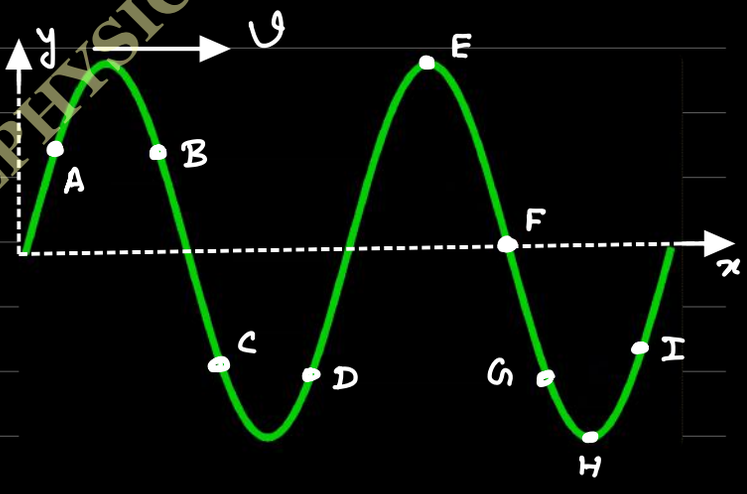
EXAMPLE: A wave is travelling to right such that disturbance at $x = 5m$ is given as $y = \frac{1}{1+t^2}$. Find the wave equation if wave velocity is 2 m/s (x, y in m, t in sec).

Sol: Applying future shift appropriately: $y = f(t - (x - x_0)/v)$

$$\Rightarrow y = \frac{1}{1+(t-(x-5)/2)^2} \Rightarrow y = \frac{4}{4+(2t-x+5)^2} \text{ Ans.}$$

→ Note that alternate form of travelling wave can also be written as $y = f(t - x/v)$ where $f(t)$ is disturbance at $x = 0$. Similarly left travelling wave can be written as $f(t + x/v)$.

EXAMPLE: Consider a string wave travelling to the right whose snapshot is shown in adjacent figure. Identify the points having velocities: (a) upwards (b) downwards (c) zero



Solution: To visualise future position of any particle we must look at configuration of particle to its immediate left (because configurations are being passed on from left to right) Similarly for knowing past configuration we need to look at configuration of particle to immediate right. Now for instance for particle A, the left particle is lower and right particle is higher thus A is moving downward with passage of time at this instant. Using this logic we can easily identify the following:

(a) Upward velocity: B, C, F, G

(b) Downward velocity: A, D, I

Now if we look at particle E, here the right particle is lower than E and left particle is also lower thus particle E was lower in past and going to be lower in immediate future, thus

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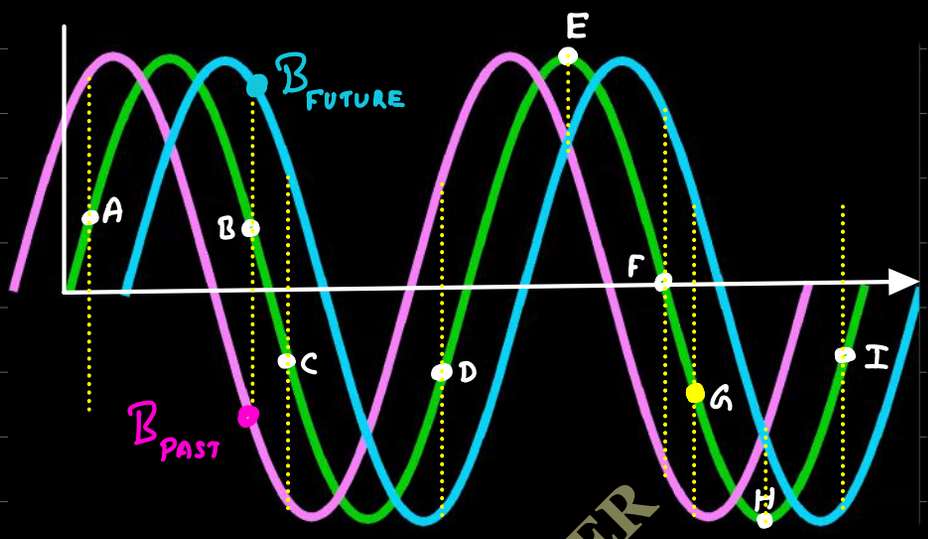
it must be at point of reversal. Thus we can identify zero velocity points as:

(i) Zero velocity: E, H.

All this can be easily seen in the adjacent figure. The future shape of the string is shown with the turquoise curve and past shape is shown with the pink curve.

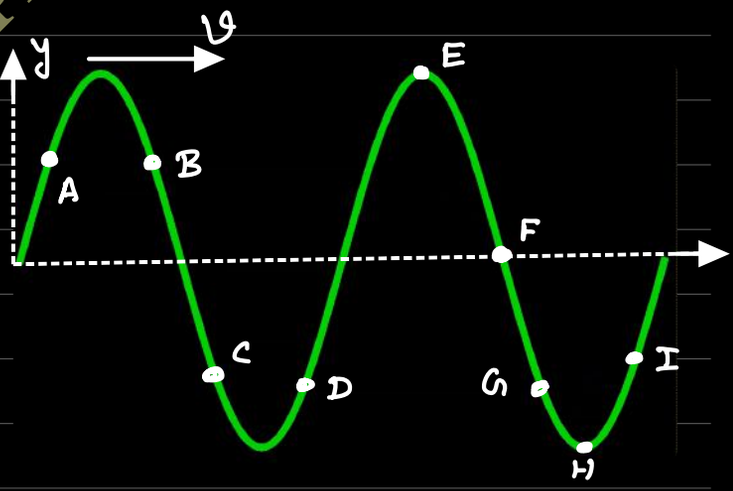


→ If the wave were travelling towards left then pink curve would represent future and turquoise curve would represent the past and all velocities will be reversed.

Finding Particle Velocity From Wave Equation:

Let wave equation be $y = f(x - vt)$ where v is signed velocity of the wave. Now particle velocity:

$$v_p = \frac{\partial y}{\partial t} = -v f'(x - vt) \quad (1)$$



here f' represents derivative of f wrt entire argument $(x - vt)$.

Similarly the slope of the wave is given by: $\frac{\partial y}{\partial x} = f'(x - vt)$ (2)

From (1) & (2), $\frac{\partial y}{\partial t} = -v \frac{\partial y}{\partial x}$ or $\boxed{v_p = -v \frac{\partial y}{\partial x}}$

Thus wave velocity can be readily inferred readily by directly looking at slope and sign of velocity (+ve for right travelling wave). Thus in our case since $v > 0$, for particles where slope is positive particle velocity is downwards and places where slope is negative, the particle velocity is upwards and at zero slope points, wave velocity is zero.

Thus: Upward: BCFG, Downwards: ADI, Zero: E, H.

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Differential Equation Governing A One Dimensional Wave:

Consider a one dimensional wave $y = f(x - vt)$

thus $\frac{\partial y}{\partial t} = -v f'(x-vt)$, $\frac{\partial y}{\partial x} = f'(x-vt)$

$\frac{\partial^2 y}{\partial t^2} = +v^2 f''(x-vt)$, $\frac{\partial^2 y}{\partial x^2} = f''(x-vt)$

thus $\boxed{\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}}$ ← One dimensional wave equation.

It can be easily seen that this differential equation also satisfies a function of the form $y = g(x+vt)$.

If $y = g(x+vt)$

$\frac{\partial y}{\partial t} = v g'(x+vt)$ $\frac{\partial y}{\partial x} = g'(x+vt)$

$\frac{\partial^2 y}{\partial t^2} = v^2 g''(x+vt)$ $\frac{\partial^2 y}{\partial x^2} = g''(x+vt)$

thus $\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}$. It can be seen that general solution of the wave differential equation is given as:

$$\boxed{y = f(x-vt) + g(x+vt)}$$

→ If $g = 0$ ⇒ Right travelling progressive wave

→ If $f = 0$ ⇒ Left travelling progressive wave

→ If $f = g$ ⇒ Stationary wave

→ If $f \neq 0, g \neq 0$, General one dimensional wave.

→ $\frac{\partial^2 y}{\partial t^2}$ represents acceleration of the particles and $\frac{\partial^2 y}{\partial x^2}$ is related to concavity of the wave shape. If concave upwards, $\frac{\partial^2 y}{\partial x^2}$ is +ve and if concave downwards, $\frac{\partial^2 y}{\partial x^2}$ is negative. Thus we can comment on acceleration of particles by just looking at concavity of the wave graph. Concave up means positive acceleration (upwards) and concave down means a negative acceleration.

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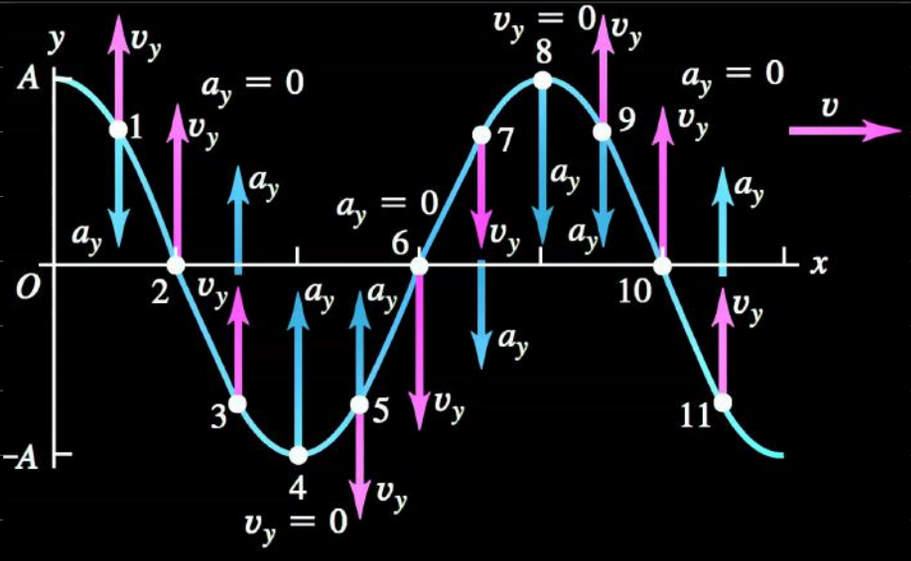
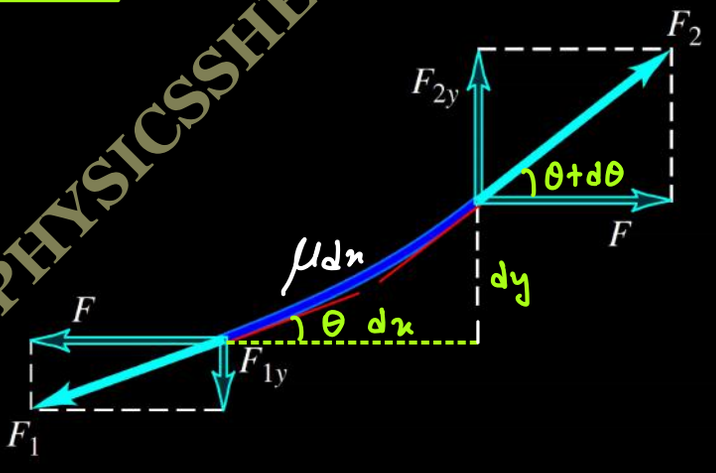
(downward acceleration).

→ For instants velocities and accelerations for the sinusoidal wave $y = A\sin(kx - \omega t)$ is as shown in adjacent figure. at $t = 0$, for all particles above x axis concavity is downwards and for all particles below x axis, concavity is upwards.



Velocity Of Transverse Wave In Elastic String:

If the motion of particles is only in up and down direction, it means horizontal acceleration of particles in the transverse wave must be zero and therefore horizontal component of tension throughout must be zero. Let this component be F then the vertical component of tension must be $\tan\theta$ at a point where the the string is inclined at an angle θ to the horizontal.



Now let mass per unit horizontal length of the string be μ then, making $\vec{F} = m\vec{a}$ equation for an element of length dx:

$$F\tan(\theta+d\theta) - F\tan\theta = \mu dx \frac{\partial^2 y}{\partial t^2}$$

$$\Rightarrow \frac{F}{dx}\left[\left.\frac{\partial y}{\partial x}\right|_{x+dx} - \left.\frac{\partial y}{\partial x}\right|_{x}\right] = \mu\frac{\partial^2 y}{\partial t^2} = F\frac{\partial^2 y}{\partial x^2}$$

$$\Rightarrow \frac{\partial^2 y}{\partial t^2} = \frac{F}{\mu}\frac{\partial^2 y}{\partial x^2}$$ now we can compare it with standard

partial differential equation governing one dimensional wave:

$\frac{\partial^2 y}{\partial t^2} = v^2\frac{\partial^2 y}{\partial x^2}$, by comparison we get wave velocity as:

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$$v = \sqrt{\frac{F}{\mu}}$$

$\mu \rightarrow$ Mass / unit length

$F \rightarrow$ Horizontal component of tension.

(For small slope wave $\cos\theta \approx 1$, thus $F \approx T$ where T is the local tension in the wave so we can also approximately write:

$$v = \sqrt{\frac{T}{\mu}} \quad (1)$$

$T \rightarrow$ Tension in the string.

EXAMPLE: A block hangs from ceiling with help of a rope having mass per unit length μ. If transverse wave pulse is generated near the block, how long will it take the pulse to reach the ceiling? Also find the acceleration of the pulse.

Solution: Consider sectional FBD of the system until a distance x measured from lower end of the rope:

$$T = mg + \mu x g = \mu v^2 \quad (\because \text{Equation (1) above}).$$

$$\Rightarrow \frac{1}{2}v^2 = \frac{1}{2}gx + \frac{m}{2\mu} \Rightarrow a = \frac{d}{dx}\left(\frac{1}{2}v^2\right) = g/2. = \text{Constant!}$$

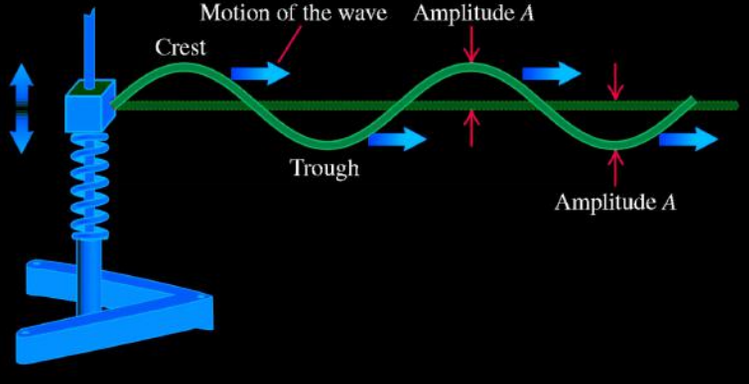
Thus pulse moves upwards with constant acceleration $\boxed{g/2}$.

Now velocity at bottom $u = \sqrt{\frac{mg}{\mu}}$, velocity at top $= \sqrt{\frac{mg + \mu l g}{\mu}} = v$

Now $T = l/v_{avg} = \frac{l\sqrt{\mu}}{2\left(\sqrt{mg} + \sqrt{mg + \mu l g}\right)}$ Ans.

SINUSOIDAL PROGRESSIVE WAVES:

Consider a simple harmonic oscillator located at $x = 0$, acting as a source for propagation of a simple harmonic disturbance with a speed v.



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Let the disturbance at $x=0$ be given by $y = A\sin(\omega t + \phi)$ then at a general x it will reach after a time delay of $t_0 = x/v$. Thus disturbance at a general x will be given by:

$$y = A\sin\left(\omega\left(t - \frac{x}{v}\right) + \phi\right) \text{ or}$$

$$y = A\sin\left(\omega t - \frac{\omega}{v}x + \phi\right), \text{ now if we put } \omega/v = k,$$

we get $y = A\sin(\omega t - kx + \phi)$, this is a sinusoidal wave with a wave velocity ω/k. Here the nomenclature of terms is as follows:

- A: Amplitude
- ω: Angular frequency
- k: Wave propagation constant
- ϕ: Phase constant
- y: Displacement from mean position
- x: x-coordinate of particle
- t: time.

In general any wave function of the form:

$$y = A\sin(\pm kx \pm \omega t \pm \phi) \text{ or } y = A\cos(\pm kx \pm \omega t \pm \phi)$$

represents a simple harmonic wave.

Time Period Of A Simple Harmonic Wave: Each particle on a string carrying a simple harmonic wave undergoes a simple harmonic motion. The time period of oscillation of each particle on the string is also called the time period of wave. Recall from SHM that time period is related to ω as:

$$\boxed{T = 2\pi/\omega}$$

This is also the time period of simple harmonic wave.

Frequency Of A Wave: Number of cycles each particle of simple harmonic wave executes per unit time is called wave frequency. Thus just like

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in SHM, here also frequency is related to time period and angular frequency as:

$$\nu = 1/T = \omega/2\pi \quad \text{or} \quad \omega = 2\pi\nu$$

WAVELENGTH: The distance through which a wave advances during one complete time period is called wavelength (λ) mathematically:

$$\lambda = vT = v/\nu$$

also since $v = \omega/k$, we can also prove $k = 2\pi/\lambda$, this is how:

$$\lambda = v/\nu = \frac{\omega/k}{\omega/2\pi} = 2\pi/k$$

$\Rightarrow k = 2\pi/\lambda$.

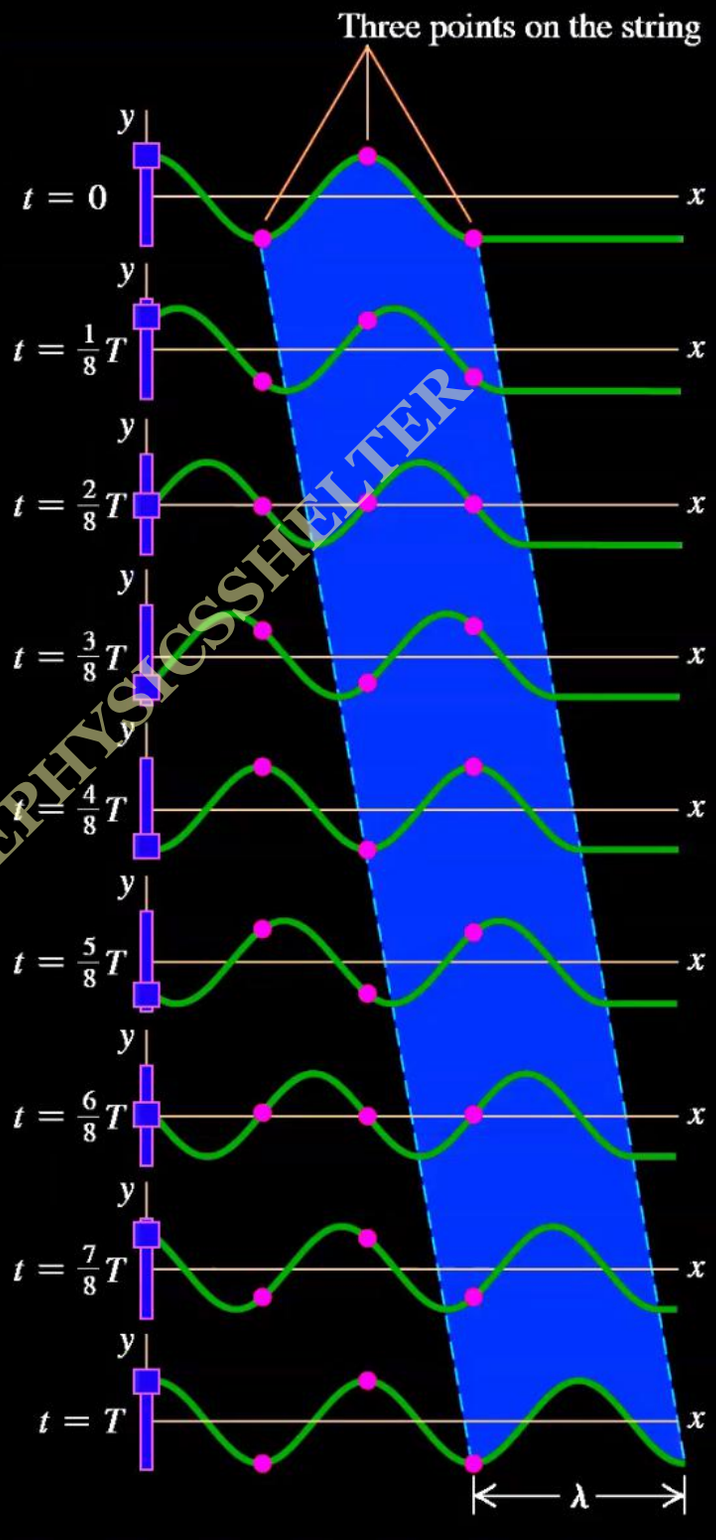
k is known as wave propagation constant and also "angular wave number".

$1/\lambda$ is called wave number.

We can also easily show that λ is equal to distance between particles in same phase (say trough to trough or crest to crest or any other phase of our choice).

For example consider a wave of the form $y = A\cos(\omega t - kx)$.

At $t = 0$, the wave shape becomes $y = A\cos(-kx) = A\cos(kx)$. For crests $\cos kx = 1 \Rightarrow kx = 0 + 2n\pi$ or $x = \frac{2n\pi}{k}$ i.e $x = 0, \frac{2\pi}{k}, \frac{4\pi}{k}$ etc.. but $k = \frac{2\pi}{\lambda}$ thus $\frac{2\pi}{k} = \lambda$



or we can say for crests, $x = 0, \lambda, 2\lambda \ldots$ etc thus λ also represents minimum separation between the particles of the string oscillating in same phase.

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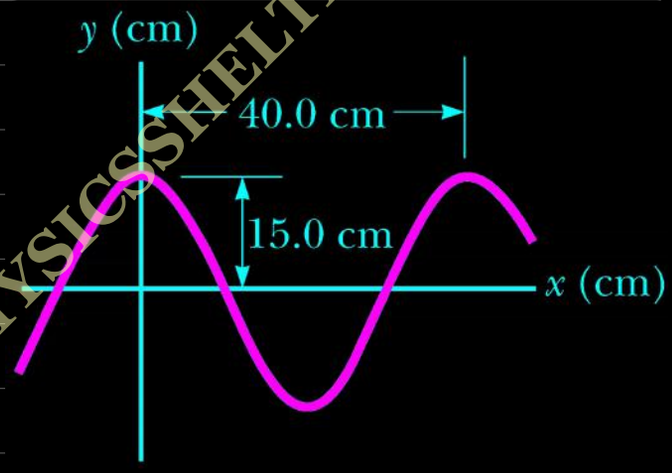
→ Using the above information, we can write a wave of the form $y = A\sin(kx - \omega t)$ as $y = A\sin\frac{2\pi}{\lambda}(x - vt)$ because, $\frac{2\pi}{\lambda} = k$ and

$2\pi v/\lambda = 2\pi\nu = \omega$. We can also write it as $y = A\sin 2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)$ we should know these alternative forms to quickly identify wavelength, time period velocity and angular frequency of the wave.

EXAMPLE:

A sinusoidal wave traveling in the positive x direction has an amplitude of 15.0 cm, a wavelength of 40.0 cm, and a frequency of 8.00 Hz. The vertical position of an element of the medium at $t = 0$ and $x = 0$ is also 15.0 cm as shown.
(A) Find the wave number k, period T, angular frequency ω, and speed v of the wave.
(B) Determine the phase constant ϕ and write a general expression for the wave function.



Assume the wave to be of the form:
$y = A\sin(kx - \omega t + \phi)$.

Solution: λ = crust to crust distance
thus $\lambda = 40\,cm \Rightarrow k = \frac{2\pi}{\lambda} = 5\pi\ rad/m$

Given $\nu = 8\,Hz \Rightarrow T = 1/\nu = 1/8$ sec
$\omega = 2\pi\nu = 16\pi\ rad\,s^{-1}$, now:
$v = \frac{\lambda}{T} = \frac{\omega}{k} = 0.4 \times 8 = 3.2\ m\,s^{-1}$

Now wave is of the form $y = A\sin(kx - \omega t + \phi)$ i.e:
$y = 0.15\sin(5\pi x - 16\pi t + \phi)$ m. Now @ $t=0$, $x=0$, $y = 0.15$ m

$\Rightarrow 0.15\sin\phi = 0.15$ or $\sin\phi = 1$ or $\phi = \pi/2$ thus our required wave:
$y = 0.15\sin(5\pi x - 16\pi t + \pi/2)$ Ans.
or $y = 0.15\cos(5\pi x - 16\pi t)$ Ans.

Think Yourself: What is the maximum particle speeds and particle acceleration of the above wave and where are the corresponding particles located.
Sol: $y = A\cos(kx - \omega t) \Rightarrow v_p = \frac{\partial y}{\partial t} = A\omega\sin(kx - \omega t)$, thus maximum particle speed is $A\omega$ i.e $0.15 \times 16\pi = 2.4\pi\ m\,s^{-1}$, this occurs for particles at mean position i.e $\sin(kx - \omega t) = \pm 1$

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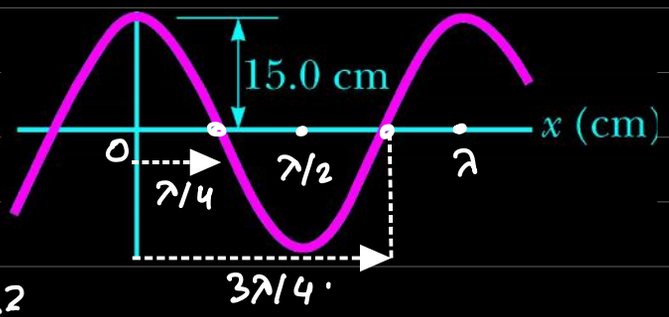
Thus at $t=0$, $\sin kx = \pm 1$ or $kx = 2n\pi \pm \pi/2$ or

$x = n \times \frac{2\pi}{k} \pm \frac{\pi}{2k} = n\lambda \pm \lambda/4$ i.e at $\frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}$ etc.



Now particle acceleration: $a_p = \frac{\partial v_p}{\partial t}$

$\Rightarrow a_p = -A\omega^2 \cos(kx - \omega t)$

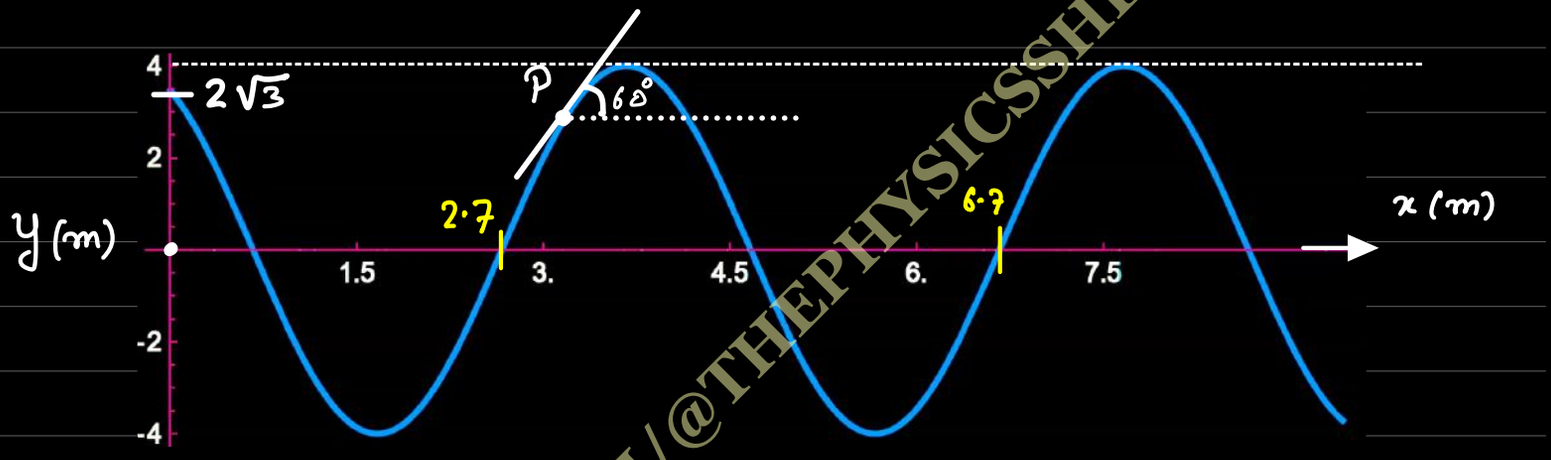
Thus $|a_{max}| = A\omega^2 = 0.15 \times (16\pi)^2$

$\Rightarrow a_{max} = 38.4\pi^2 \, ms^{-2}$.

This must occur extreme positions of wave particles. For a_{max}, $\cos kx = \pm 1$

$\Rightarrow kx = n\pi$ or $x = n\lambda/2$ i.e $0, \lambda/2, \lambda$ etc.

EXAMPLE:



The figure shows snapshot of a vibrating string at $t=0$. The particle P is observed to be moving up with a velocity of $20\sqrt{3} \, ms^{-1}$. The tangent at P makes an angle $60°$ with x axis. Find:

(a) The direction in which the wave is travelling.

(b) The wave equation.

Solution: $v_p = -v(\partial y/\partial x) \Rightarrow +20\sqrt{3} = -v \tan 60° = -v\sqrt{3}$

$\Rightarrow v = -20 \, ms^{-1}$. Thus wave is moving leftwards at $20 \, ms^{-1}$.

Now $\lambda = 6.7 - 2.7 = 4m$. Now $k = 2\pi/\lambda = \pi/2$, $\omega = vk = 10\pi$

Amplitude $A = 4m$. Now let the snapshot be $y = A\sin(kx + \phi)$,

at $x=0$, $y = A\sin\phi = 2\sqrt{3} = 4\sin\phi \Rightarrow \sin\phi = \frac{\sqrt{3}}{2}$, also $\frac{dy}{dx}$ is negative, thus $A\&\cos\phi$ must be negative, thus,

$\phi = 2\pi/3$. $\Rightarrow$ @ $t=0$, $y = 4\sin\left(\frac{\pi}{2}x + \frac{2\pi}{3}\right)$, now at a general time it will be left shifted by $20t$.

Thus required wave equation: $y = 4\sin\left(\frac{\pi}{2}(x + 20t) + \frac{2\pi}{3}\right)$

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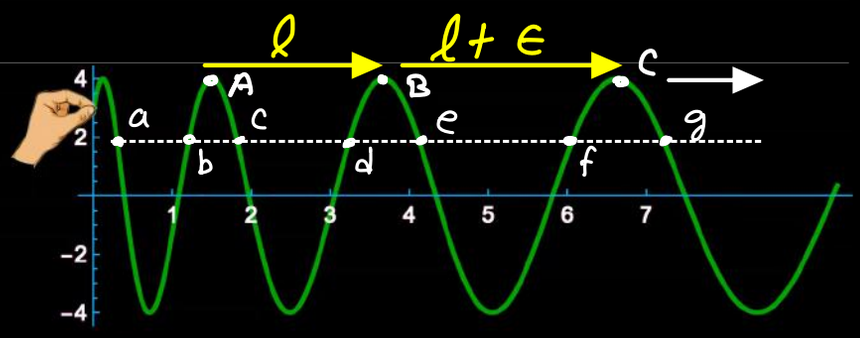
Thus required wave equation $y = 4\sin\left(\frac{\pi}{2}x + 10\pi t + \frac{2\pi}{3}\right)$ cms.

EXAMPLE:



A boy is oscillating his hand simple harmonically in a string with same horizontal tension throughout. He creates a wave travelling to right as shown. (a) What could be possible reason for crests becoming farther apart as we move away from the boy.
(b) At the shown instant which particle out of a, b, c, d, e, f and g have maximum and minimum speeds.

Solution: The frequency of oscillation of each particle must be same as the frequency of oscillation of the source (boy's hand) as we are moving away the wavelength of the wave is increasing (distance between particles oscillating in same phase is wavelength). Now $v = \lambda\nu$, however since ν is constant therefore speed must be increasing. Now $v = \sqrt{T/\mu}$ since T is same throughout, increase in velocity of wave must mean that μ must be decreasing, the the string is having lesser and lesser mass per unit length as we move away from the boy. (This could be due to either string tapering off or due to gradually changing material). (The wave is accelerating)

(b) All the particle a, b, c, d, e, f and g are executing SHM of same amplitude and frequency and are equidistant from mean position. The speed of each particle must be $\omega\sqrt{A^2 - y^2}$ since y coordinate is same for all of them thus each one of them has same speed.

Think yourself: If you were given length l and $l + \varepsilon$ and time period of oscillation, could you find acceleration of the wave (assumed constant).

Solution: Crest located at B would have travelled a time T extra as compared to that at A and crest located at C

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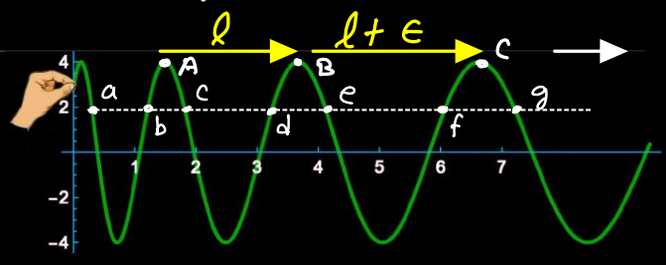
would have travelled a time $2T$ extra as compared to crest at A. (T is time period of oscillation).



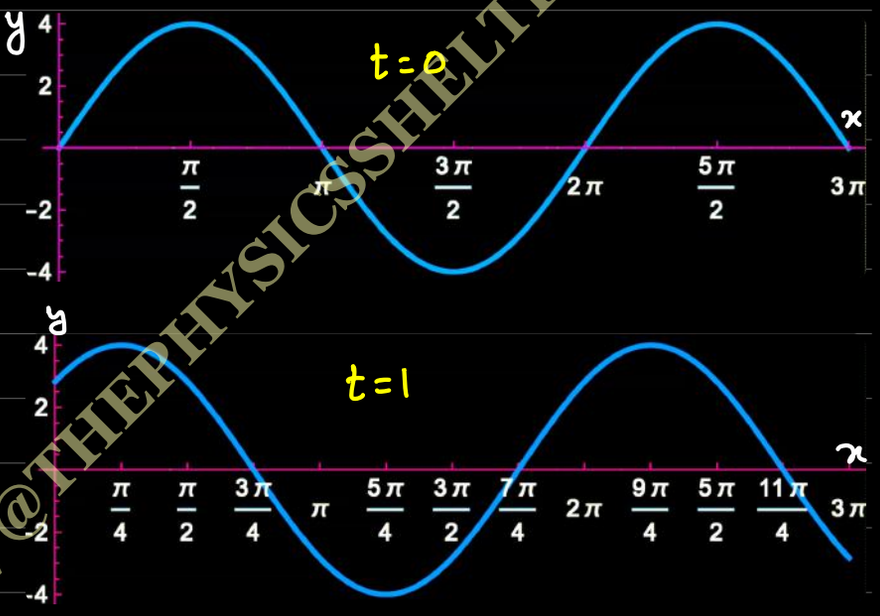
Let wave speed at A be u and acceleration be a then by kinematic of 1D uniformly accelerated motion:

$$\ell = uT + \frac{1}{2}aT^2 \quad (1)$$

$$2\ell + \epsilon = u(2T) + \frac{1}{2}a(2T)^2 \quad (2)$$

Solving (1) and (2): $a = \epsilon/T^2$ and $u = (2\ell - \epsilon)/2T$ Ans.

EXAMPLE: The figures show snapshots of a string carrying a sinusoidal waves at $t=0$ and $t=1$ sec respectively. Find the possible family of wave equations that satisfies the given situation. All given data is in SI units. (String is infinite, only a portion is shown here).



Solution: From first figure we can see that $\lambda = 2\pi$ metres.

From figure 2 we can see that intercept at π has possibly shifted leftwards to $\frac{3\pi}{4}$ (or an additional multiple of λ).

Thus displacement of wave is $\left(\frac{3\pi}{4} - \pi\right) + n\lambda$ where n could be a positive or negative integer. Thus wave velocity: $\left(-\frac{\pi}{4} + n \times 2\pi\right) \div 1$

Now $t=0$, wave shape is $y = 4\sin(x)$, thus at a general time, wave shape $= 4\sin(x - vt) = 4\sin\left(x - \left(2n\pi - \frac{\pi}{4}\right)t\right)$ Ans.

Note that wave could be travelling right as well as left.

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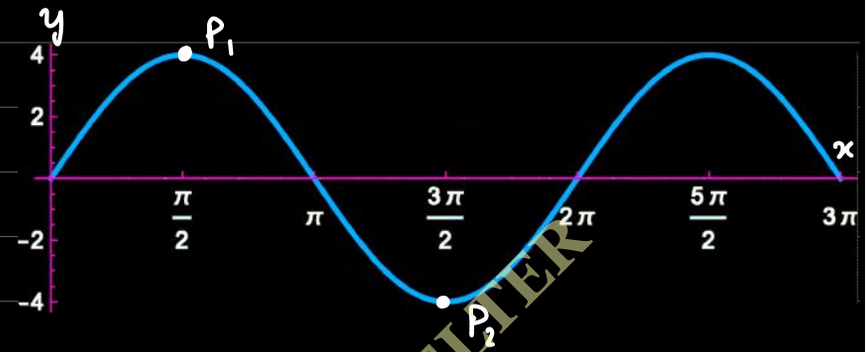
Phase Of A Wave:

If we have a sinusoidal wave of the form $y = A\sin(kx \pm \omega t \pm \phi)$ or $A\cos(kx \pm \omega t \pm \phi)$ then the entire argument of the sine or cosine term is called the phase of the wave.

Phase Difference Between Two Particles At A Given Time:



If we have a wave of the form $y = A\sin(\omega t - kx)$ then for finding the phase difference between two particles at a given time, we should first make the k positive. Thus we can write $y = A\sin(kx - \omega t + \pi)$

Now suppose at some time t_0 we are looking at two particles located at x_2 and x_1, we have: $\phi_2 = kx_2 - \omega t_0 + \pi$ and $\phi_1 = kx_1 - \omega t_0 + \pi$, now phase difference is given as simply:

$$\Delta\phi = \phi_2 - \phi_1 = k(x_2 - x_1) = k\Delta x, \quad \text{thus:}$$

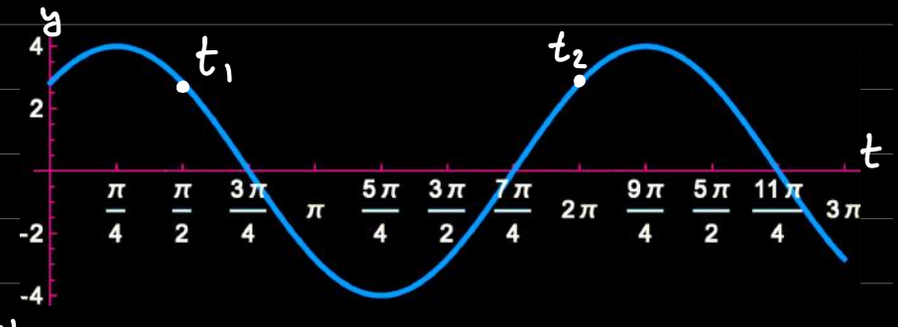
$$\boxed{\Delta\phi = \frac{2\pi}{\lambda}\Delta x}$$

For example for the wave shown above,

phase difference between particles P_2 and P_1 is $\frac{2\pi}{\lambda} \times \left(\frac{3\pi}{2} - \frac{\pi}{2}\right)$

$\Rightarrow \quad \Delta\phi = \pi$

Phase Difference For A Given Particle At Two Different Times:



If we have a wave function of the form $y = A\sin(kx - \omega t)$ then to find the phase difference we must first make ω positive i.e write $y = A\sin(\omega t - kx + \pi)$

Now for a particle at $x = x_0$, the phase at two times t_1 and t_2 are given as $\phi_2 = \omega t_2 - kx_0 + \pi$ and $\phi_1 = \omega t_1 - kx_0 + \pi$.

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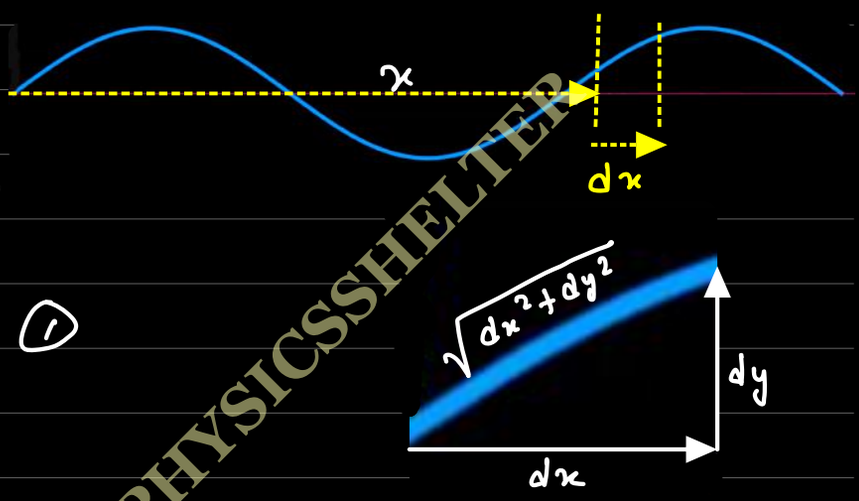
Now phase difference for the particle at two times is given as $\Delta\phi = \phi_2 - \phi_1 = \omega(t_2 - t_1) = \omega\Delta t = \frac{2\pi}{T}\Delta t$. Thus:

$$\boxed{\Delta\phi = \frac{2\pi}{T}\Delta t}$$

For instance, for the shown graph, phase difference between times t_2 and t_1 is given as $\frac{2\pi}{T}\left(2\pi - \frac{\pi}{2}\right) = \frac{3\pi}{2}$ $(\because T = 2\pi)$

Energy Per Unit Length Of A Wave:

Consider an element of thickness dx on a string having a mass per unit length μ.



Now kinetic energy of this element must be:

$$dKE = \frac{1}{2}\mu dx\left(\frac{\partial y}{\partial t}\right)^2 \quad (1)$$

$(\because dm = \mu dx \;\&\; v_p = \partial y/\partial t)$.

Now the stretched length of the string segment is given as: $dl = \sqrt{(dx)^2 + (dy)^2}$ while its natural length was dx. Now let the tension in the string be T then elastic energy stored in the string must be $T\,dl$ (Think of an already stretched spring having a tension kx, then $dU = kx\,dx$ $(\because U = \frac{1}{2}kx^2)$) Note that $T\,dl$ is elastic energy stored over and above the initial energy stored due to tension). Thus,

$$dPE = T\left[\sqrt{(dx)^2 + (dy)^2} - dx\right]$$

$$\Rightarrow dPE = T dx\left(\sqrt{1 + (dy/dx)^2} - 1\right)$$

Now if we have a small slope wave then $dy/dx << 1$ and we can apply binomial approximation, thus:

$$dPE = T dx\left[\frac{1}{2}\left(\frac{dy}{dx}\right)^2\right] = \frac{1}{2}\mu v^2\left(\frac{\partial y}{\partial x}\right)^2 dx \quad \left(\because v = \sqrt{\frac{T}{\mu}}\right) \quad (2)$$

(Since y is a function of x and time, by dy/dx we meant $\partial y/\partial x$).

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From ① & ②, Total Energy per unit length:

$$\frac{dE}{dx} = \frac{dKE}{dx} + \frac{dPE}{dx} = \frac{1}{2}\mu\left[\left(\frac{\partial y}{\partial t}\right)^2 + v^2\left(\frac{\partial y}{\partial x}\right)^2\right]$$

thus $$\boxed{\frac{dE}{dx} = e_l = \frac{1}{2}\mu\left[\left(\frac{\partial y}{\partial t}\right)^2 + v^2\left(\frac{\partial y}{\partial x}\right)^2\right]}$$

or $e_l = \frac{1}{2}\mu G$ where $\boxed{G = \left(\frac{\partial y}{\partial t}\right)^2 + v^2\left(\frac{\partial y}{\partial x}\right)^2}$

we have defined G because we will come across this factor in several other expressions also.

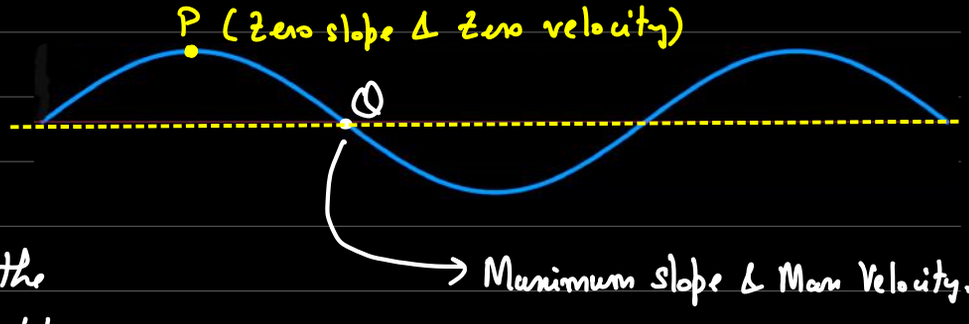
KINETIC ENERGY & POTENTIAL ENERGY EQUALITY IN PROGRESSIVE WAVES:

From previous section we have seen that KE/unit length:

$\frac{dKE}{dx} = \frac{1}{2}\mu\left(\frac{\partial y}{\partial t}\right)^2$ and potential energy per unit length:

$\frac{dPE}{dx} = \frac{1}{2}\mu v^2\left(\frac{\partial y}{\partial x}\right)^2$, However for travelling waves recall the relation between particle velocity and wave velocity:

$\frac{\partial y}{\partial t} = -v\frac{\partial y}{\partial x}$, thus we can clearly see that kinetic energy per unit length in any segment must be equal to potential energy per unit length in that segment for a progressive wave (travelling wave).



This might come as surprise to some students however realise that potential energy is related to stretching in the string and maximum stretching occurs at the point of maximum slope, which is also the mean position of the particle. Thus at mean position both KE & PE are maximum while at extreme position both are zero and always equal in any segment of a travelling wave.

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This might look like a violation of energy conservation to students however, since the segment is connected to its previous segment so it can recieve energy from its previous segment and can also pass on energy to the next segment, therefore it would be theoretically unsound to apply energy conservation on any segment of the string and thus is absolutely no violation of conservation of mechanical energy.

Average Energy Per Unit Length For A Sinusoidal Wave:

To find the average energy per unit length for a sinusoidal wave we will need the value of $\langle \cos^2 \omega t \rangle$ and $\langle \sin^2 \omega t \rangle$ where angular brackets represent average value of quantity in the bracket. It can be easily shown that $\langle \sin^2 \omega t \rangle = \langle \cos^2 \omega t \rangle = 1/2$. We know that

$$\sin^2 \omega t + \cos^2 \omega t = 1 \quad \text{thus} \quad \langle \sin^2 \omega t + \cos^2 \omega t \rangle = 1$$

Also $\langle \sin^2 \omega t + \cos^2 \omega t \rangle = \langle \sin^2 \omega t \rangle + \langle \cos^2 \omega t \rangle = 1$ and by symmetry $\langle \sin^2 \omega t \rangle = \langle \cos^2 \omega t \rangle$ thus we have the required result:

$$\boxed{\langle \sin^2 \omega t \rangle = \langle \cos^2 \omega t \rangle = 1/2.}$$

Now recall that $\rho_e = \frac{1}{2} \mu G$ where $G = \left(\frac{\partial y}{\partial t}\right)^2 + v^2 \left(\frac{\partial y}{\partial x}\right)^2$

now suppose $y = A \sin(kx - \omega t)$ then $\frac{\partial y}{\partial t} = -A\omega \cos(kx - \omega t)$

and $\frac{\partial y}{\partial x} = Ak \cos(kx - \omega t)$, thus:

$$G = A^2 \left[\omega^2 \cos^2(kx - \omega t) + v^2 k^2 \cos^2(kx - \omega t) \right]$$

but $v = \omega/k$, thus:

$$G = 2A^2 \omega^2 \cos^2(kx - \omega t) \Rightarrow \langle G \rangle = A^2 \omega^2 \quad (\because \langle \cos^2 \omega t \rangle = 1/2)$$

Thus average energy per unit length for a sinusoidal wave:

$$\boxed{\langle \rho_e \rangle = \frac{1}{2} \mu A^2 \omega^2}$$

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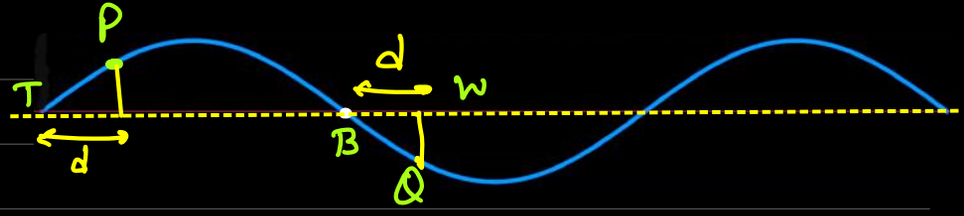
→ Total energy of a segment can be obtained by multiplying length of segment by average energy only if the length is exact multiple of half wavelengths because it ensures that all types of elements are included. For instance when we consider the energy between P and Q, it will be same as energy between T and B (We left out TP but included BQ which by symmetry has same energy as TP section).



EXAMPLE: A certain sinusoidal wave is given as $y = A\sin(kx - \omega t)$ Find the energy stored in the wave @ $t = 0$ from $x = 0$ to $x = \frac{\lambda}{3}$ given mass per unit length is μ.

Solution: Since kinetic energy and potential energy of every segment in a progressive wave are equal, we can just find any one of them and double it to find the total energy.

Thus our required expression is:

$$\int_0^{\lambda/3} \frac{1}{2}\mu\left(\frac{\partial y}{\partial t}\right)^2 dx \times 2 \Bigg|_{t=0}$$

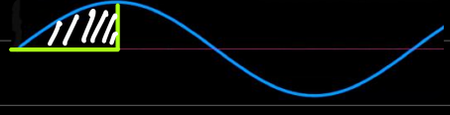
Now $\partial y/\partial t = -A\omega\cos(kx - \omega t) = -A\omega\cos kx$ @ $t = 0$

Thus our required integral is:

$$\mu A^2\omega^2 \int_0^{\lambda/3} \cos^2\left(\frac{2\pi}{\lambda}\right) x\, dx = \frac{\mu A^2\omega^2}{2}\left[\frac{1}{3} - \frac{\sqrt{3}}{8\pi}\right] \quad \text{Ans.}$$

→ If we have to find total energy in a large number of wavelengths, then also we can use average energy to get approximately correct results.

→ We can also use average energy if we wish to find energy in perfect quarter loops.



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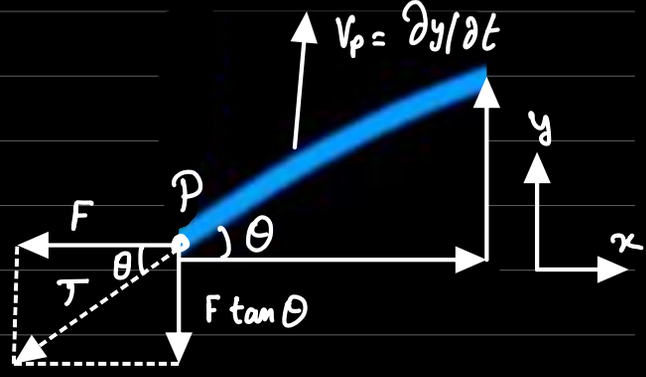
Power Transmission In A Wave:

Consider a point P on a string. Let horizontal component of tension throughout the string be F. Now refer to adjacent figure, we know that Power,

$$P = \vec{F} \cdot \vec{v} = F\tan\theta(-\hat{j}) \cdot \left(\frac{\partial y}{\partial t}\right)\hat{j}$$



$$\Rightarrow \boxed{P = -F\frac{\partial y}{\partial x}\cdot\frac{\partial y}{\partial t}}$$, Now if we have a small slope wave,

$F = T\cos\theta \approx T$ thus we can also write:

$$\boxed{P = -T\frac{\partial y}{\partial x}\cdot\frac{\partial y}{\partial t}}$$

→ This expression is valid for all kinds of one dimensional waves on string (progressive or stationary or a mixture).

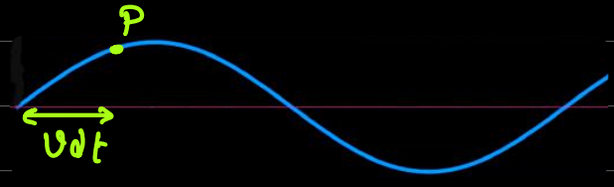
→ For progressive waves $\frac{\partial y}{\partial t} = -v\frac{\partial y}{\partial x}$ thus we can also write power as: $P = Tv\left(\frac{\partial y}{\partial x}\right)^2 = \mu v\left[v^2\left(\frac{\partial y}{\partial x}\right)^2\right]$ $(\because T = \mu v^2)$

Thus $P = \frac{1}{2}\mu v\left[\left(\frac{\partial y}{\partial t}\right)^2 + v^2\left(\frac{\partial y}{\partial x}\right)^2\right] = \frac{1}{2}\mu v G.$

thus for progressive waves, $\boxed{P = \frac{1}{2}\mu v G}$ ← Only for Progressive Waves.

Alternative Derivation For Power Of Progressive Waves:

Consider a point P on the string. The wave configuration travels a distance vdt in a time dt thus energy stored in a segment of length vdt just before this point must have crossed the point P. Thus, if dE is the energy crossed:



$$dE = \frac{1}{2}\mu G \times v\,dt \Rightarrow \frac{dE}{dt} = \frac{1}{2}\mu v G$$

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This is the same result that we obtained above for Power thus power in a travelling (or progressive) wave is:

$$\boxed{P = \frac{1}{2}\mu v \omega^2} = \boxed{\frac{1}{2}\mu v A^2\omega^2}$$ for sinusoidal progressive waves

← Average

EXAMPLE: DAMPENING WAVE: A pulse travelling along a string of linear mass density μ is described by the wave function:

$y = [A_0 e^{-bx}]\sin(kx - \omega t)$ where the factor in bracket can be considered to be amplitude. (a) Find Avg. power carried by this wave $P(x)$. (b) What is the power carried at the origin $P(0)$. (c) Compute the ratio $P(x)/P(0)$. [$P(x)$, $P(0)$ represent average power values].

Solution: Strictly speaking the function does not represent a travelling wave, however if b is sufficiently small, over a small distance we can treat it like a sinusoidal wave of nearly constant amplitude. Now recall that for a sinusoidal wave, power $= \frac{1}{2}\mu v A^2\omega^2$, thus for our case,

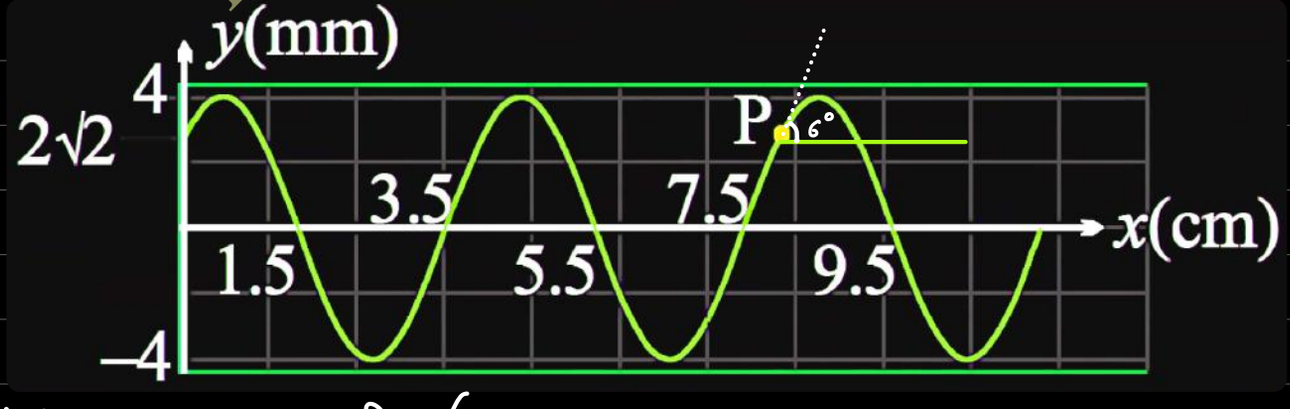
$$P_{avg}(x) = \frac{1}{2}\mu A_0^2 e^{-2bx}\cdot\left(\frac{\omega}{k}\right)\omega^2 = \frac{\mu\omega^3}{2k}A_0^2 e^{-2bx} \text{ Ans.}$$

Now $P_{avg}(0) = \frac{\mu\omega^3}{2k}A_0^2$ Ans. $\frac{P(x)}{P(0)} = e^{-2bx}$ Ans.

EXAMPLE:

The figure shows a snap photograph of a vibrating string at $t = 0$. The particle P is observed moving up with speed 20π cm/s. The tangent at P makes an angle $6°$ with the x-axis. Take $\mu = 50$ gm/m. [Note the unequal units on x axis & y axis]

Find the instantaneous power at point P.



Solution: $P = -T\frac{\partial y}{\partial t}\left(\frac{\partial y}{\partial x}\right)$, Here $\frac{\partial y}{\partial t} = 0.2\pi \text{ ms}^{-1}$

Now since $6°$ is the angle when y is in mm and x

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is in cm, thus if the graph were plotted with equal scales, the angle would become $0.6°$ [Note that for small angles, $\tan\theta \approx \theta$].

Also, $T = \mu v^2$ and $v_p = -v\left(\frac{\partial y}{\partial x}\right) \Rightarrow T = \mu\left(\frac{\partial y/\partial t}{\partial y/\partial x}\right)^2$,

thus power: $P = -\mu\left(\frac{\partial y/\partial t}{\partial y/\partial x}\right)^2 \times \left(\frac{\partial y}{\partial t}\right)\left(\frac{\partial y}{\partial x}\right) = -\mu\left(\frac{\partial y}{\partial t}\right)^3 \Big/ \left(\frac{\partial y}{\partial x}\right)$

Putting in the values: $P = \frac{-0.05 \times (0.2\pi)^3}{0.6 \times \pi/180} \approx -1.18\,\mu W$

Intensity Of String Waves: Power trasmitted per unit cross sectional area of the string is called Intensity. Thus:

$$I = \frac{P}{S} = \frac{1}{2}\frac{\mu v G}{S}$$

$dm = \mu dx = \rho S dx$, S (dx) $\Rightarrow \rho = \mu/S$

$\Rightarrow$ $\boxed{I = \frac{1}{2}\rho v G}$ (S is the cross sectional area of the string).

Average Intesity Of Sinusoidal Progressive Waves:

$I_{avg} = \frac{1}{2}\rho v \langle G \rangle = \frac{1}{2}\rho v A^2\omega^2$ or alternatively,

$I_{avg} = \langle P \rangle / S = \frac{1}{2}\mu v A^2\omega^2 / S = \frac{1}{2}\rho v A^2\omega^2$, thus:

$$\boxed{I_{avg} = \frac{1}{2}\rho v A^2\omega^2}$$

Average Power Of Superposition Of Two Sinusoidal Waves (Antiparallel):

Let the wave function of a wave travelling in a string be:

$$y = A\sin(kx - \omega t + \phi_1) + B\sin(kx + \omega t + \phi_2)$$

Now $P = -T\frac{\partial y}{\partial x}\cdot\frac{\partial y}{\partial t} = -Tk\omega\left[-A^2\cos^2(kx - \omega t + \phi_1) + B^2\cos^2(kx + \omega t + \phi_2)\right]$ (1)

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$$\Rightarrow \langle P \rangle = -Tk\omega \left[-\frac{A^2}{2} + \frac{B^2}{2} \right] = \frac{Tk\omega}{2} \left[A^2 - B^2 \right]$$

Now let P_1 and P_2 be the individual powers of component waves:
$y_1 = A\sin(kx - \omega t + \phi_1)$ and $y_2 = B\sin(kx + \omega t + \phi_2)$.

Now $P_1 = -T \frac{\partial y_1}{\partial t} \cdot \frac{\partial y_1}{\partial x} = A^2 kT\omega \cos^2(kx - \omega t + \phi_1)$ ②

$$P_2 = -T \frac{\partial y_2}{\partial t} \cdot \frac{\partial y_2}{\partial x} = -B^2 kT\omega \cos^2(kx + \omega t + \phi_2)$$ ③

From ①, ② and ③ it can be clearly seen that:

$$\boxed{P = P_1 + P_2} \text{ and } \boxed{\langle P \rangle = \langle P_1 \rangle + \langle P_2 \rangle}$$

Thus power of superposition of two oppositely directed sinusoidal waves is simple superposition of powers of individual component travelling waves.

Power Of Superposition Of Two Sinusoidal Waves: (Parallel Velocities)

Consider a wave $y = y_1 + y_2 = A\sin(kx - \omega t + \phi_1) + B\sin(kx - \omega t + \phi_2)$

Now:

$$P = -T \frac{\partial y}{\partial x} \cdot \frac{\partial y}{\partial t} = Tk\omega \left[A^2\cos^2(kx - \omega t + \phi_1) + B^2\cos^2(kx - \omega t + \phi_2) + 2AB\cos(kx - \omega t + \phi_1)\cos(kx - \omega t + \phi_2) \right]$$ ①

Now if we only have $y_1 = A\sin(kx - \omega t + \phi_1)$,

$$P_1 = -T \frac{\partial y_1}{\partial x} \cdot \frac{\partial y_1}{\partial t} = Tk\omega A^2\cos^2(kx - \omega t + \phi_1)$$ ②

Similarly $P_2 = Tk\omega B^2\cos^2(kx - \omega t + \phi_2)$ ③

From ①, ② & ③, $\boxed{P = P_1 + P_2 + 2\sqrt{P_1 P_2}}$

Now the product term $2AB\cos(kx - \omega t + \phi_1)\cos(kx - \omega t + \phi_2)$ can be written as: $AB\cos(\phi_1 - \phi_2) + AB\cos(2kx - 2\omega t + \phi_1 + \phi_2)$

Its average value is simply $AB\cos(\phi_1 - \phi_2)$, (∵ the other

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terms is a simple phasor whose average value is zero).
Thus using equation ①, Average power:

$$\langle P \rangle = T\omega h \left[A^2/2 + B^2/2 + AB\cos(\phi_1 - \phi_2) \right]$$

$$\Rightarrow \boxed{\langle P \rangle = \langle P_1 \rangle + \langle P_2 \rangle + 2\sqrt{\langle P_1 \rangle \langle P_2 \rangle}\cos(\Delta\phi)} \quad \Delta\phi = |\phi_1 - \phi_2| \quad ④$$

This could also be seen directly using: $A^2 = A_1^2 + A_2^2 + 2A_1A_2\cos(\Delta\phi)$

→ Power of a wave travelling along $+x$ is positive everywhere because $\partial y/\partial t \times \partial y/\partial x$ always have negative sign everywhere. (This simply means energy is travelling from left to right at all points).

→ Power of a left travelling wave (along $-x$) is negative everywhere (This means energy is travelling in $-x$ direction everywhere).

→ From ④ we can also write a similar equation in intensities, If I_1 and I_2 are individual average intensities due to two waves at any point then net Intensity at that point is given as:

$$\boxed{I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\Delta\phi}$$

Energy Per Unit Length Of Superposed Sinusoidal Waves (Anti Parallel):

Let a wave be superposition:

$$y = y_1 + y_2 = A\sin(kx - \omega t + \phi_1) + B\sin(kx + \omega t + \phi_2)$$

Now $e_l = \frac{1}{2}\mu\left[\left(\frac{\partial y}{\partial t}\right)^2 + v^2\left(\frac{\partial y}{\partial x}\right)^2\right]$, substituting y and using $k = \omega/v$:

$$e_l = \mu\omega^2\left[A^2\cos^2(kx - \omega t + \phi_1) + B^2\cos^2(kx + \omega t + \phi_2)\right]$$

Thus $\boxed{e_l = e_{l_1} + e_{l_2}}$

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Energy Per Unit Length Of Superposed Sinusoidal Waves (Parallel):

Let a wave be given as $y = y_1 + y_2 = A\sin(kx - \omega t + \phi_1) + B\sin(kx - \omega t + \phi_2)$

Now $e_l = \frac{1}{2}\mu\left(\left(\frac{\partial y}{\partial t}\right)^2 + v^2\left(\frac{\partial y}{\partial x}\right)^2\right) = \mu\left(\frac{\partial y}{\partial t}\right)^2$ (∵ Both terms are equal for a progressive wave).

Now substituting for y:

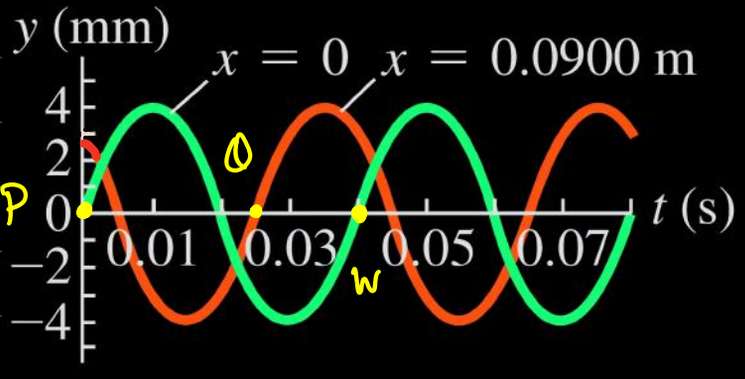
$$e_l = \mu\omega^2\left[A^2\cos^2(kx - \omega t + \phi_1) + B^2\cos^2(kx - \omega t + \phi_2) + 2AB\cos(kx - \omega t + \phi_1)\cos(kx - \omega t + \phi_2)\right]$$

$$\Rightarrow \boxed{e_l = e_{l_1} + e_{l_2} + 2\sqrt{e_{l_1} e_{l_2}}}$$

$$\rightarrow \boxed{\langle e_l \rangle = \langle e_{l_1} \rangle + \langle e_{l_2} \rangle + 2\sqrt{\langle e_{l_1} \rangle \langle e_{l_2} \rangle}\cos(\Delta\phi)}$$

→ The last result is also evident from the fact that $A^2 = A_1^2 + A_2^2 + 2A_1A_2\cos(\Delta\phi)$.

Example: A sinusoidal wave is propagating along a stretched string lying along $+x$ axis. Displacement vs time graphs are given for two particles located at $x = 0$ and $x = 0.09$ m. (a) What is the wavelength if two particles are within one wavelength of each other and wave is travelling to the right. (b) What is instantaneous power at $x = 0.09$ at $t = 0$ if tension in the string is 1N.



Solution: From graph, $T = 0.04$ sec, $A = 4$ mm, thus oscillation at $x = 0$ is given as $A\sin(\omega t)$ and at general x it is $y = A\sin\left(\omega\left(t - \frac{x}{v}\right)\right)$. Now focus on points marked P and Q on the graph, we can see that we can see that events at Q are delayed from events at P by a time difference of 0.02S sec. AMBARISH SIR

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Thus wave velocity $v = 0.09/0.025 = 3.6\ ms^{-1}$, thus $\lambda = vT$

$\Rightarrow \lambda = 3.6 \times 0.04 = 0.144\ m$ Ans.

Thus our wave equation becomes:

$$y = 4\sin\left(\frac{2\pi}{0.04}\left(t - \frac{x}{3.6}\right)\right) \times 10^{-3}\ m. = 0.004\sin\left(50\pi t - \frac{125\pi x}{9}\right)\ m.$$

Now power: $-T\left(\frac{\partial y}{\partial t}\right)\cdot\left(\frac{\partial y}{\partial x}\right) = Tv\left(\frac{\partial y}{\partial x}\right)^2 \quad \left(\because \frac{\partial y}{\partial t} = -v\frac{\partial y}{\partial x}\right)$

$\Rightarrow P = Tv\left(Ak\cos(\omega t - kx)\right)^2$ @ $t=0$, $P = Tv\left[A\times\frac{2\pi}{\lambda}\cos\left(\frac{2\pi}{\lambda}x\right)\right]^2$

Thus for our data, $P = 1 \times 3.6 \times \left[4\times10^{-3}\times\frac{2\pi}{0.144}\times\cos\left(\frac{125\pi}{9}\times 0.09\right)\right]^2$

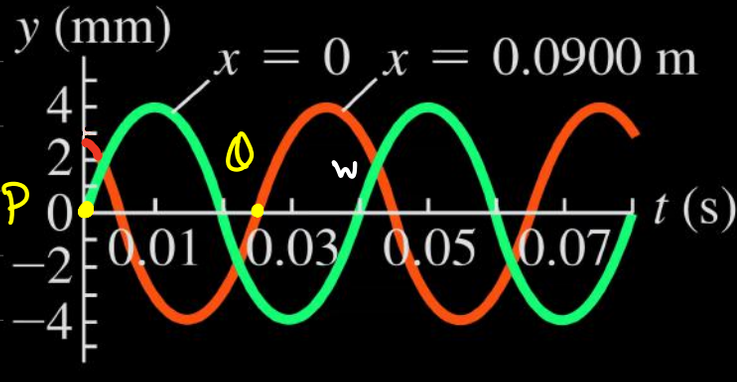
$\Rightarrow P = 0.109\ W$

THINK YOURSELF: If wave were travelling left, what would be wavelength?

Solution: Focus on points W and O, Event that occurred at $x = 0.09$ at $t = 0.025$ sec occurs at $x=0$ at time 0.04 sec, Thus, $v = 0.09/(0.04 - 0.025) = 6\ ms^{-1}$. Thus $\lambda = vT = 6\times0.04 = 0.24\ m$ Ans.

Now if we want we can also write wave equation as $y = A\sin\omega\left(t + \frac{x}{v}\right)$

THINK YOURSELF: What would be the wavelength if it were known that separation between two particles lies between λ and 2λ and wave is travelling to the right.



Solution: Event at any x is same as event at $x - \lambda$. Thus event at $x_2 = 0.09$ is same as at $x = x_1 = 0.09 - \lambda$. Now once again we can focus on points P and O thus:

$$(0.09 - \lambda) = v \times 0.025 = \frac{\lambda}{T}\times 0.025 = \frac{\lambda}{0.04}\times 0.025$$

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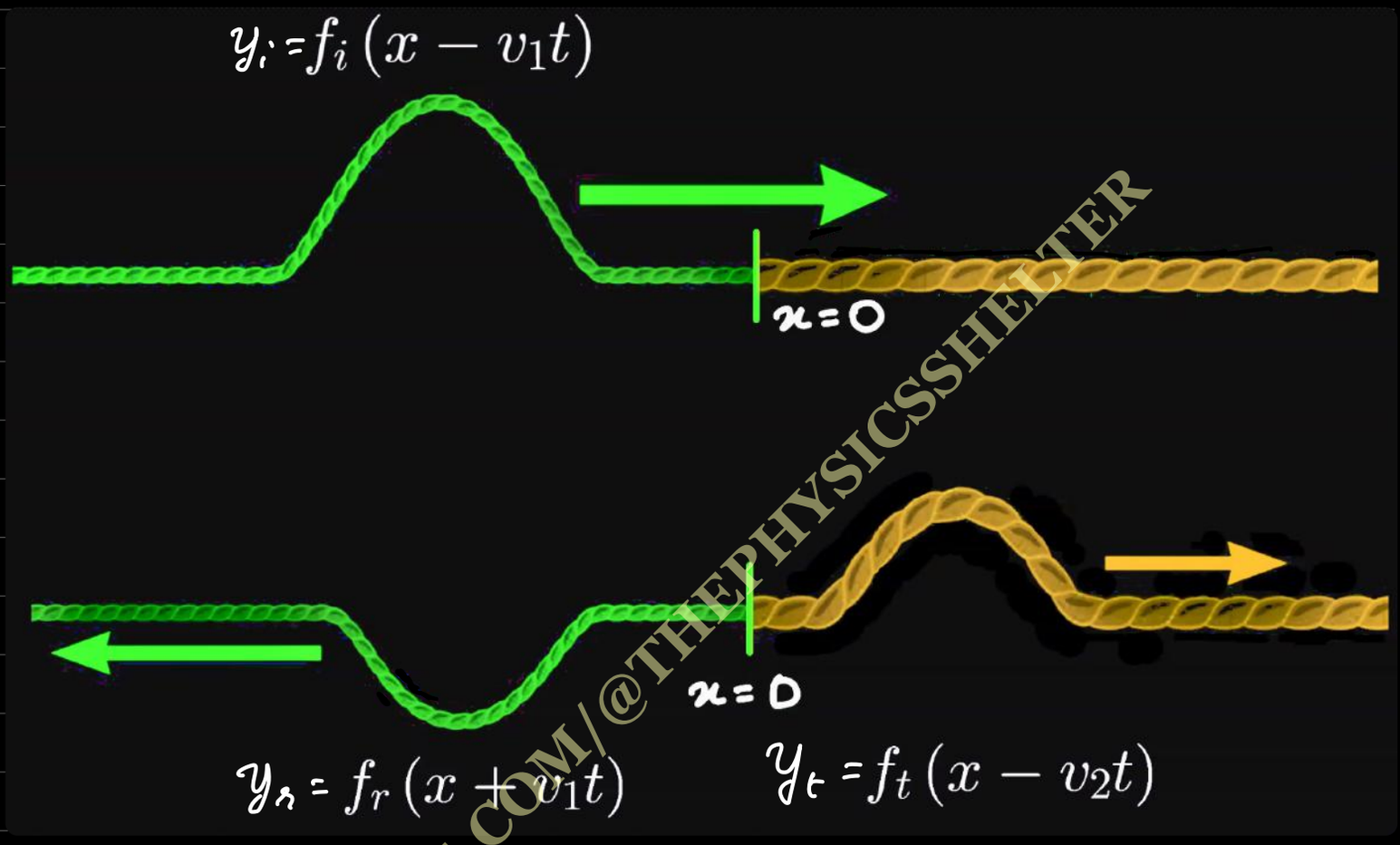
$\Rightarrow (0.09 - \lambda) = \frac{\lambda}{0.04} \times 0.025 \Rightarrow \lambda = 0.055\,m.$

We can similarly solve for λ if we knew x lying between $n\lambda$ and $(n+1)\lambda$.

Reflection And Transmission Of Waves:



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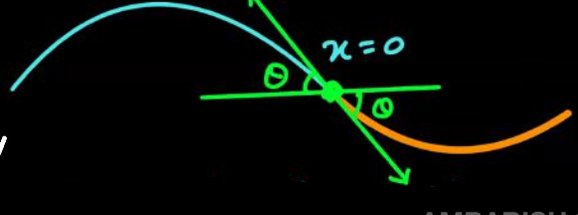
Consider a wave incident at a boundary of two strings as shown in the figure, we wish to find the equation of transmitted and reflected waves. For this we shall use the following conditions at the boundary: @ $x = 0$

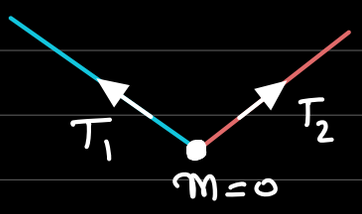
① $y_i + y_r = y_t$ ($\because$ Otherwise string will break, $\therefore$ Continuity of displacement required).

② $y_i' + y_r' = y_t'$ (Continuity of slope, otherwise boundary will have ∞ acceleration)

See adjacent figure, the boundary is a massless point, therefore tension acting on it from two strings must be equal and opposite, thus, the slopes must be same at



the boundary. If the slopes change abruptly across the boundary, the resultant force on boundary particle can't be zero.



Now differentiating equation ① at boundary:

$$\dot{y}_i + \dot{y}_r = \dot{y}_t \quad ③ \text{ [At boundary only]}.$$

Also: $y_i' = -\frac{1}{v_1}\dot{y}_i$, $y_t' = -\frac{1}{v_2}\dot{y}_t$, $y_r' = \frac{1}{v_1}\dot{y}_r$ $\left(\because \frac{\partial y}{\partial t} = -v\frac{\partial y}{\partial x}\right)$

using these substitutions in equation ②:

$$-\frac{1}{v_1}\dot{y}_i + \frac{1}{v_1}\dot{y}_r = \frac{1}{v_2}\dot{y}_t \quad ④ \quad \text{[At the boundary only]}.$$

Now equations ③ & ④ are simultaneous equations in $\dot{y}_r$ and $\dot{y}_t$ and can be readily solved in terms of $\dot{y}_i$. Solving we get:

$$\dot{y}_t = \frac{2v_2}{v_1+v_2}\dot{y}_i \quad ⑤ \text{ and } \dot{y}_r = \frac{v_2-v_1}{v_2+v_1}\dot{y}_i \quad ⑥$$

Integrating ⑤ and ⑥ we get:

$$y_t = \frac{2v_2}{v_1+v_2}y_i + C_1, \quad y_r = \frac{v_2-v_1}{v_2+v_1}y_i + C_2$$

However when nothing is incident, there is nothing to be reflected or transmitted thus if $y_i = 0$, $y_t = 0$ & $y_r = 0 \Rightarrow C_1 = C_2 = 0$

Thus we have relations for transmitted and reflected disturbances at the boundary as:

$$\boxed{y_t = \frac{2v_2}{v_1+v_2}y_i} \text{ and } \boxed{y_r = \frac{v_2-v_1}{v_2+v_1}y_i}$$

Here we took reflecting boundary at $x=0$, however our derivation is independent of the choice of location of reflecting boundary.

Steps For Finding Reflected & Transmitted Waves:

Suppose we have reflecting boundary at $x=a$ and our incident wave is $y_i = f(x-v_1t)$, then we follow these steps:

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1. Evaluate incident disturbance at $x=a$ $y_i = f(a - v_1 t)$

2. Calculate transmitted disturbance as $y_t = \frac{2v_2}{v_1+v_2} f(a - v_1 t)$

3. Calculate transmitted wave by applying a future shift of $(x-a)/v_2$ to y_t at the boundary i.e $y_t = \frac{2v_2}{v_1+v_2} \times f\left(a - v_1\left[t - \frac{(x-a)}{v_2}\right]\right)$

4. Calculate reflected disturbance at the boundary as $y_r = \frac{v_2 - v_1}{v_2+v_1} f(a - v_1 t)$

5. Calculate reflected wave by future shifting by $\left(\frac{a-x}{v_1}\right)$ i.e

$$y_r = \left(\frac{v_2 - v_1}{v_2+v_1}\right) f\left(a - v_1\left[t - \frac{(a-x)}{v_1}\right]\right)$$

Phase Relations Between Transmitted And Reflected Waves:

(Animation) Reflection From Rarer Medium ($v_2 > v_1$)

(Animation) Reflection from denser medium ($v_2 < v_1$)

Recall that $y_r = \left(\frac{v_2 - v_1}{v_2+v_1}\right) y_i$ thus if $v_2 > v_1$ then if

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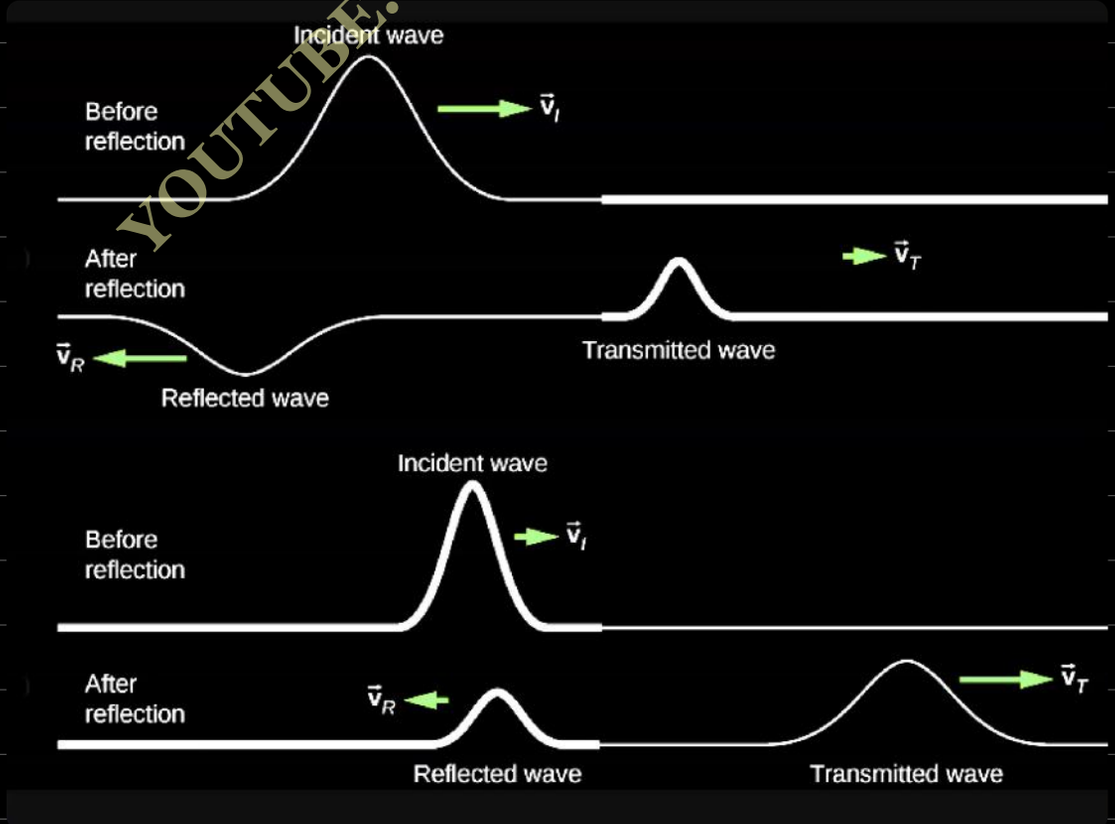
y_i is positive, y_r is also positive and thus a mountain returns as a mountain though a little diminished thus we say upon reflection from a rarer medium, there is no phase change (rarer medium means $v_2 > v_1$).

Now when $v_2 < v_1$, $\left(\frac{v_2 - v_1}{v_2 + v_1}\right)$ factor becomes negative and thus a mountain comes back as a valley (still diminished in magnitude though). Thus we say that upon reflection from a denser medium, there is a phase shift of π.

Now recall that $y_t = \frac{2v_2}{v_1 + v_2} y_i$, the factor $2v_2/v_1+v_2$ is always positive thus a mountain is always transmitted as a mountain. Further if $v_2 > v_1$ (transmission in to a rarer medium), the factor $\frac{2v_2}{v_1+v_2}$ is greater than 1 thus a mountain is transmitted as a higher mountain. Similarly if $v_2 < v_1$ (transmission in to a denser medium), the factor $2v_2/v_1+v_2 < 1$ thus a mountain is transmitted as a lower mountain.



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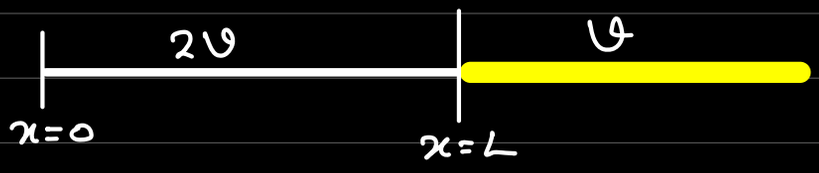
EXAMPLE:



At $x=0$ we have a source of incident wave oscillating as $y = A\sin(\omega t)$. There is a reflecting boundary at $x=L$. The wave velocity in two segments of the string are $2v$ and v as shown. Find:

(a) Incident wave $y_i(x,t)$
(b) Incident disturbance $y_i(L,t)$
(c) Transmitted disturbance $y_t(L,t)$
(d) Transmitted wave
(e) Reflected disturbance and Reflected wave $y_r(x,t)$.

Solution: (a) $y_i = A\sin\left[\omega\left(t - \frac{x}{2v}\right)\right]$ (∵ future shift $= x/2v$)

(b) $y_i(L,t) = A\sin\left[\omega t - \frac{\omega L}{2v}\right]$ Ans

(c) Now $y_t = (2v_2/v_1+v_2)\,y_i = \left(\frac{2v}{3v}\right) A\sin\left(\omega t - \frac{\omega L}{2v}\right)$

$y_t(L,t) = \frac{2}{3} A\sin\left(\omega t - \frac{\omega L}{2v}\right)$ Ans

(d) Now $y_t(x,t) = \frac{2}{3} A\sin\left(\omega\left[t - \left(\frac{x-L}{v}\right)\right] - \frac{\omega L}{2v}\right) =$

$\Rightarrow y_t(x,t) = \frac{2}{3} A\sin\left(\omega t - \frac{\omega x}{v} + \frac{\omega L}{2v}\right)$ Ans

(e) Now $y_r(L,t) = \left(\frac{v_2 - v_1}{v_2 + v_1}\right) y_i = -\frac{1}{3} A\sin\left(\omega t - \frac{\omega L}{2v}\right)$ Ans.

Now $y_r(x,t) = -\frac{1}{3} A\sin\left(\omega\left[t - \frac{(L-x)}{2v}\right] - \frac{\omega L}{2v}\right)$

$\Rightarrow y_r(x,t) = -\frac{A}{3}\sin\left(\omega t + \frac{\omega x}{2v} - \frac{\omega L}{v}\right)$ Ans.

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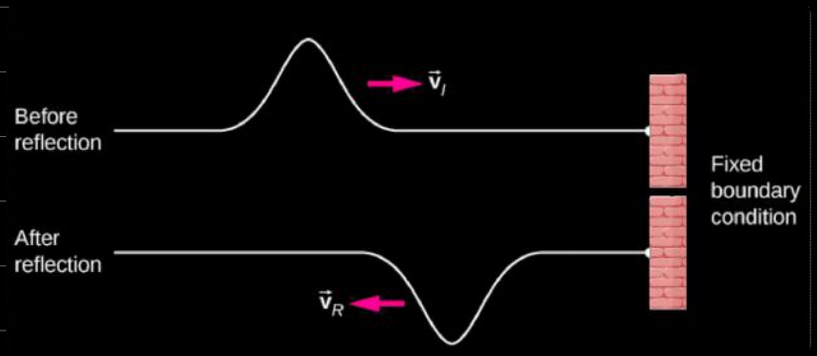
Reflection From Perfectly Rigid Boundary:



At fixed boundary, there can be no displacement thus, $y_i + y_r = 0$

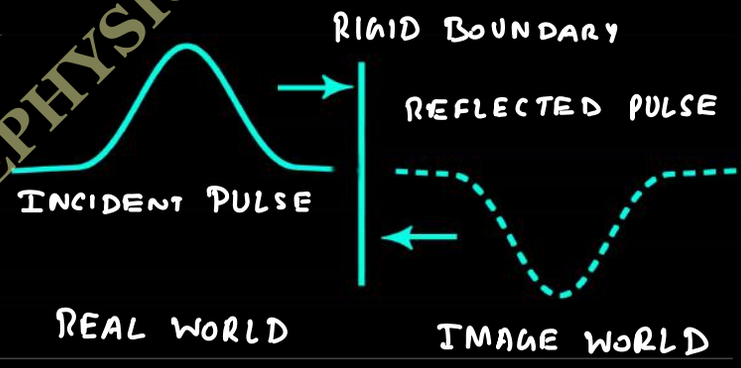
$\Rightarrow \boxed{y_r = -y_i}$. Thus a mountain returns as a valley of same shape and size.

[A rigid boundary can also be thought of as a string of infinite mass density, thus $v_2 = 0$,

Now using our formula derived in previous section:

$$y_r = \left(\frac{v_2 - v_1}{v_2 + v_1}\right) y_i = -y_i \qquad \text{(This is the same result)}$$

Geometrical Modelling Of Reflection From Fixed End (Rigid Boundary)



From our analysis in previous section we can model the reflected pulse as mirror image of the water image of the incident of pulse and the net shape of the string will be superposition of the two pulses.

This will be more clearly understood from the following example:

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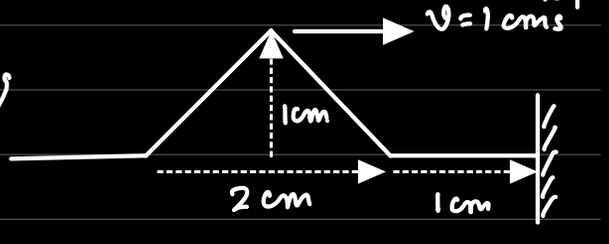
Example:

Consider a symmetric triangular pulse approaching a rigid boundary as shown in the figure.

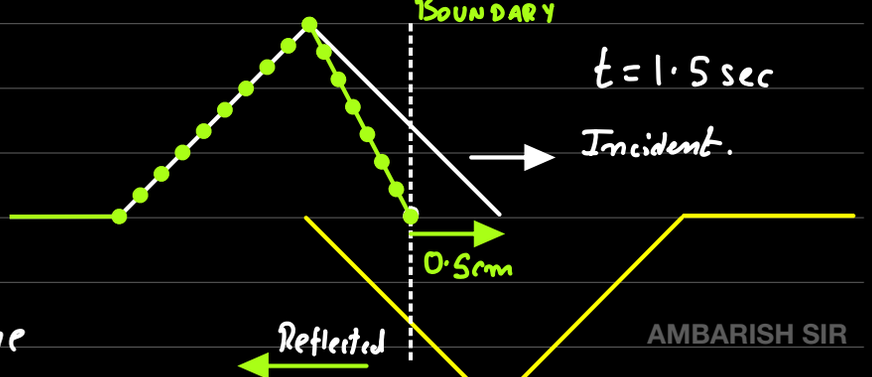
Sketch the shape of the string at:

(a) 1.5 sec (b) 2 sec (c) 2.5 sec (d) 3 Sec



Solution: (a) The adjacent figure shows the situation at $t = 1.5$ sec

Both the reflected pulse and incident pulse have advanced by 1.5 cm. The green dotted line



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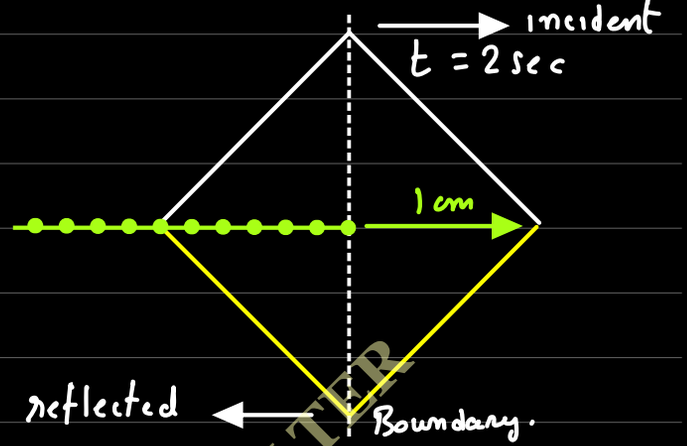
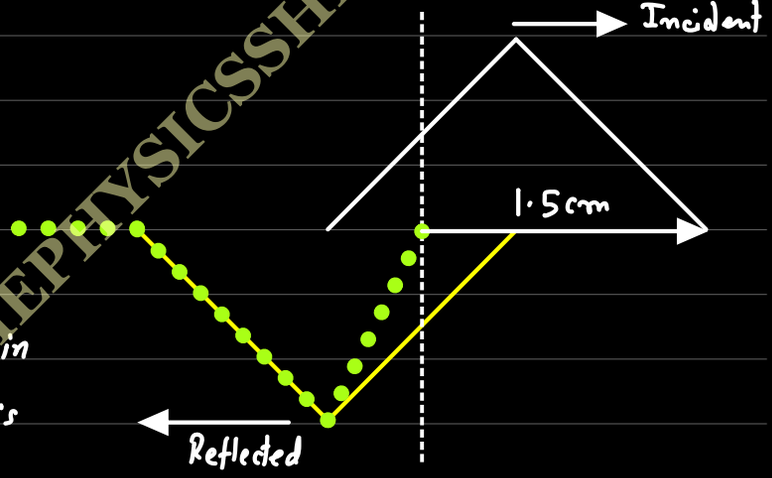
shows the actual shape of the string. (Note that superposition of linear functions is a straight line.). The green line simply represents the addition of the two functions (incident function in white colour and reflected function in yellow colour).

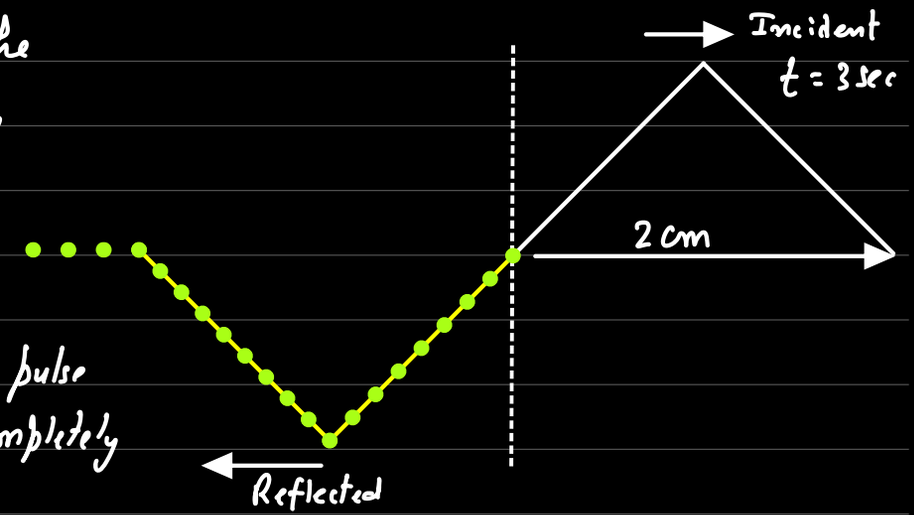
(b) The adjacent figure shows the situation at $t = 2$ sec. The dotted green line represents the shape of the string. At this instant entire energy is in the form of kinetic energy (Note that all particles have downward velocity due to incident as well as reflected waves).



(c) The adjacent figure shows the situation at 2.5 sec. The green dotted line shows the actual shape of the string. A portion of kinetic energy is once again converted to potential energy by this instant.



(d) The adjacent figure shows the situation at $t = 3$ seconds, by this time incident pulse has completely passed in to image world while reflected pulse is completely in real world. The pulse that came as mountain is completely returning as a valley.



→ Note that the pulse in this example can't represent a real wave due to sharp change in slope (discontinuous slope) and it is taken for only academic convenience to understand superposition of incident and reflected waves.

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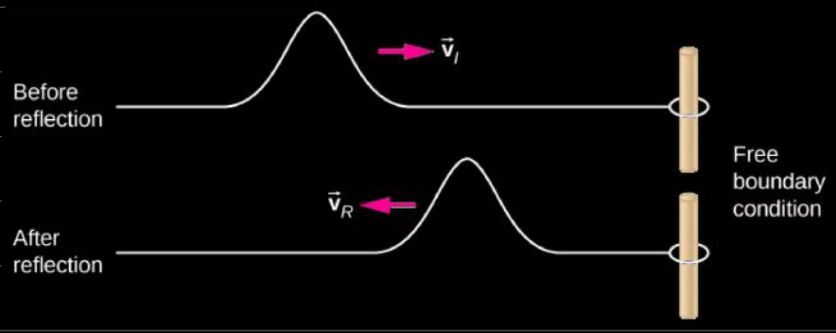
Reflection From Free Boundary:

A free boundary is modelled as end of a string attached to a massless ring free to slide in a frictionless rod. Since the ring is massless therefore it can't have any vertical component of tension acting on it. Therefore, the slope of the string at this boundary must always be zero. Thus:

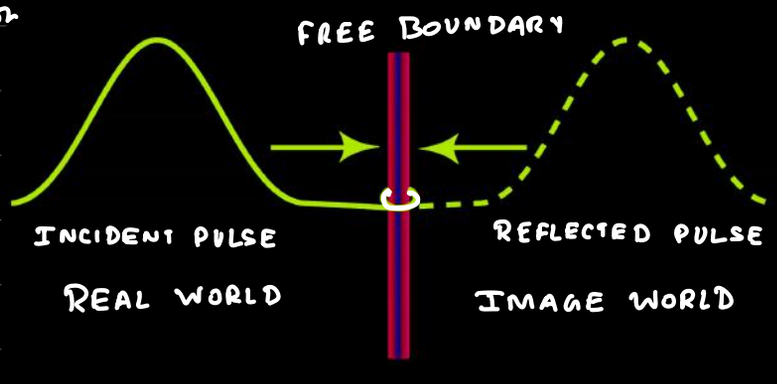


$$y_i' + y_r' = 0 \Rightarrow \frac{-1}{v}\dot{y}_i + \frac{1}{v}\dot{y}_r = 0 \quad \left(\because \frac{\partial y}{\partial t} = -v\frac{\partial y}{\partial x}\right)$$

thus: $\dot{y}_i = \dot{y}_r \Rightarrow y_i = y_r + C$, but when nothing is incident, there is nothing to be reflected thus $C = 0$ therefore at reflecting boundary, $\boxed{y_i = y_r}$, thus a mountain returns as a mountain at the free boundary. (Alternatively a free boundary is interpret as being connected to a massless string, thus $v_2 = \infty$ (infinitely rare medium). Thus according to our formula for reflected disturbance, $y_r = \left(\frac{v_2 - v_1}{v_2 + v_1}\right) y_i = y_i$ or $y_r = y_i$, this is the same result that we derived above.

Geometrical Modelling From A Free Boundary:

From our discussing from previous section we can geometrically model the reflected pulse simply as a mirror reflection of incident pulse in the reflecting boundary. Now the net shape of the string can be obtained by superposition of incident pulse and reflected pulse. This will be clearer from the following example.



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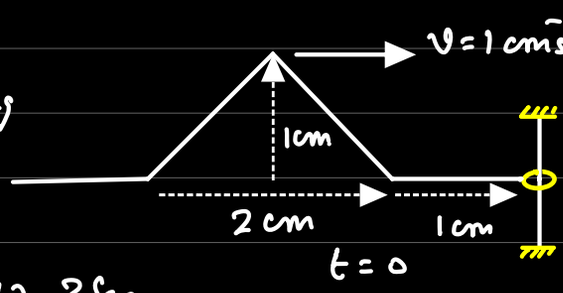
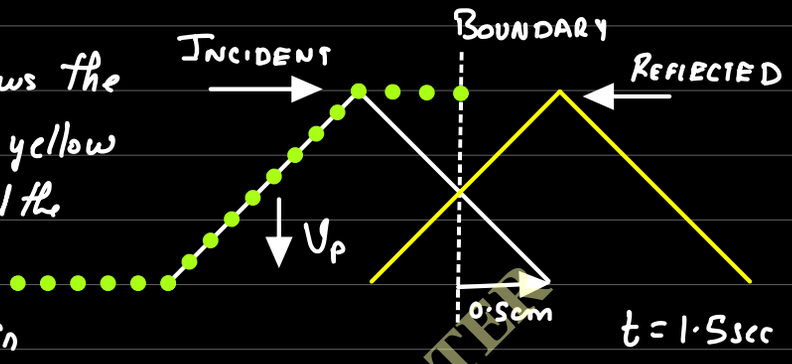
EXAMPLE:

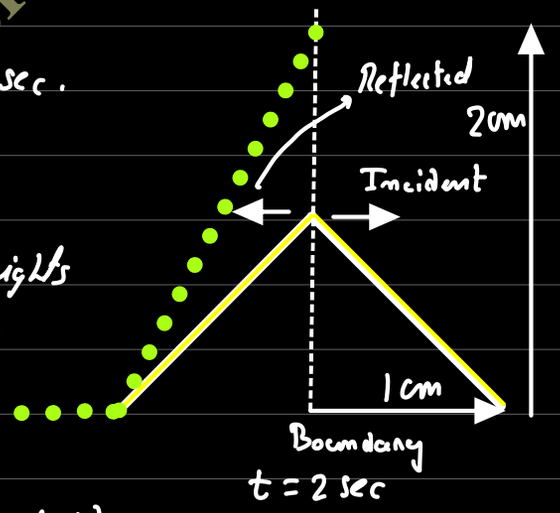
Consider a symmetric triangular pulse approaching a free boundary as shown in the figure. Sketch the shape of the string at:
(a) 1·5 sec (b) 2 sec (c) 2·5 sec (d) 3 Sec



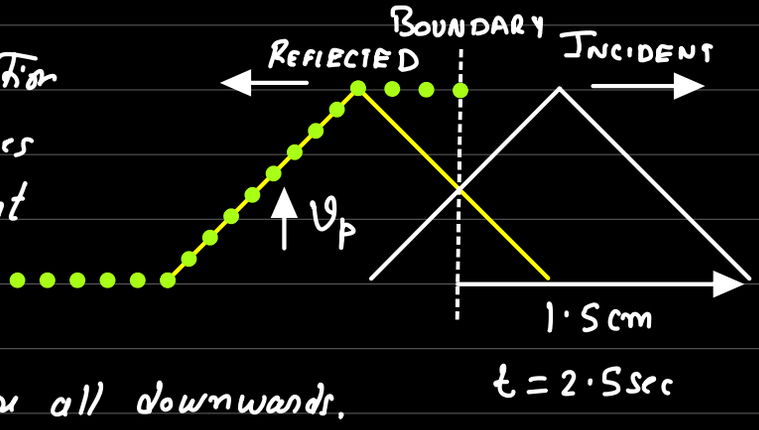
Solution: (a) The adjacent figure shows the situation at $t = 1.5$ seconds. The yellow pulse represents reflected pulse and the white pulse is the incident one. The net shape of the string is their superposition and shown by the dotted green lines. Note that net slope at the boundary is zero because at the boundary slopes of incident and reflected pulses are negative of each other.



(b) Adjacent figure shows the situation at $t = 2$ sec. At this time reflected and incident pulses exactly overlaps the reflected pulse and maximum height achieved is double the maximum heights of both the pulses. At this instant entire energy is in the form of potential energy and the kinetic energy is zero. (Note that velocities due to both pulses are oppositely directed).



(c) The adjacent figure shows the situation at 2·5 seconds. The dotted lines represent the superposition of incident (White) and reflected (Yellow) pulses. Note that velocities of particles on slanted portion of the string are all downwards.



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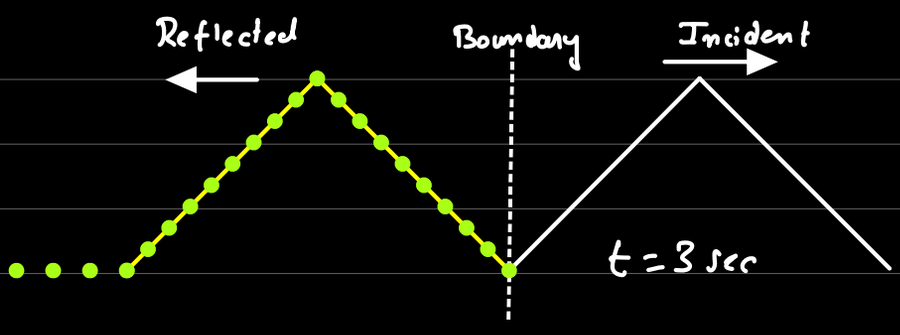
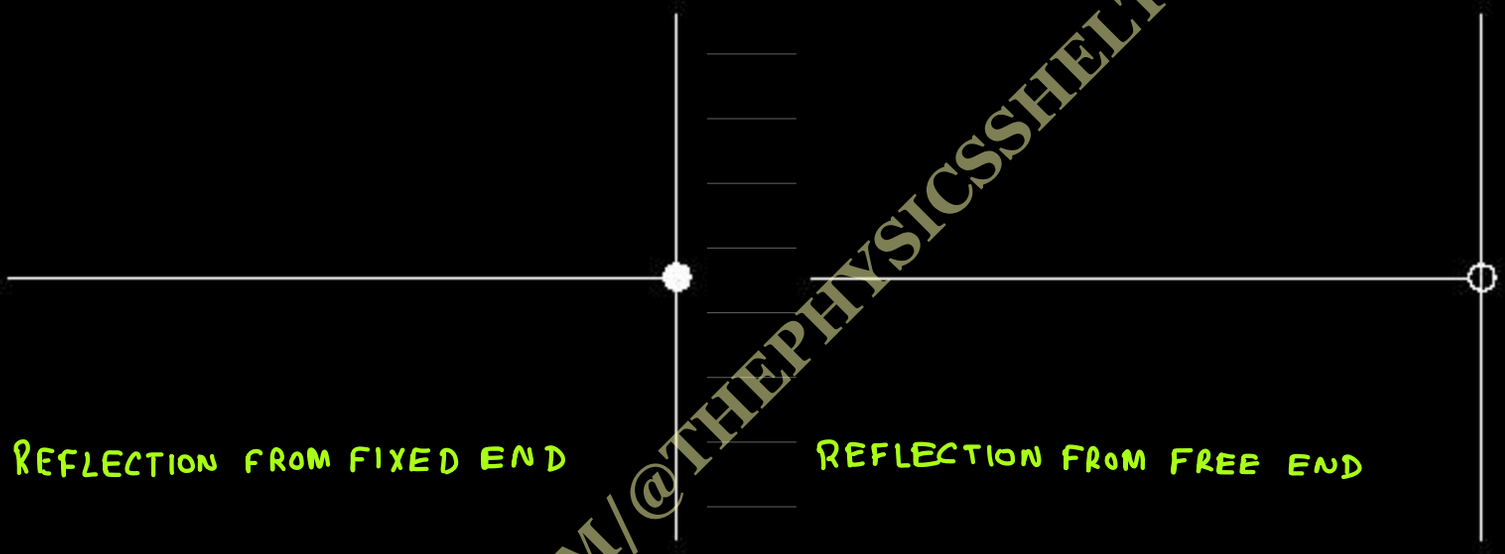
(d) Adjacent figure shows situation at $t = 3$ sec. The reflected pulse has completely entered real world and incident pulse has completely passes in to image world. Here we see how a mountain incident on a free boundary returns as a mountain.



Animations For Fixed Boundary & Free Boundary Reflections:



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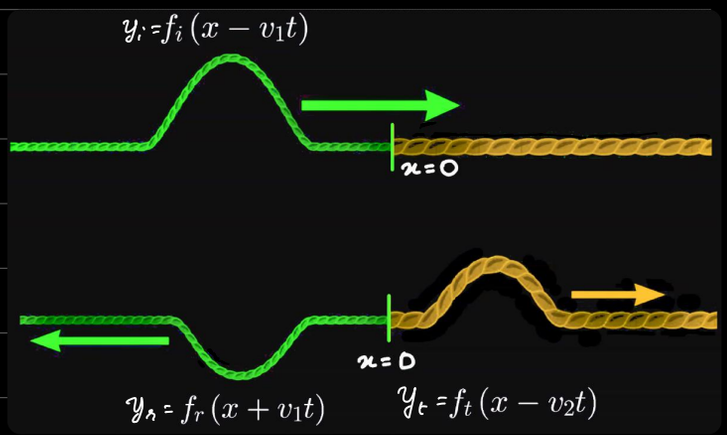
Energy Transmission And Reflection Across A Boundary:

Recall that power of a progressin wave is given as:

$$P = \frac{1}{2}\mu v\left[\left(\frac{\partial y}{\partial t}\right)^2 + v^2\left(\frac{\partial y}{\partial x}\right)^2\right]$$

$$= \mu v\left(\frac{\partial y}{\partial t}\right)^2 = \mu v \dot{y}^2$$



Thus incident power $= \mu_1 v_1 \dot{y}_i^2$

Reflected power $= \mu_1 v_1 \dot{y}_r^2$

Thus reflected fraction: $\eta = (\dot{y}_r / \dot{y}_i)^2$

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Now recall that at the boundary, $y_r = \left(\frac{v_2 - v_1}{v_2 + v_1}\right) y_i$.

$\Rightarrow y_r = \left(\frac{v_2 - v_1}{v_2 + v_1}\right) y_i$. Thus $\eta = \left(\frac{v_2 - v_1}{v_2 + v_1}\right)^2$, Thus reflected energy fraction formula is given as:

$$\boxed{\eta_r = \left(\frac{v_2 - v_1}{v_2 + v_1}\right)^2}$$

Now transmitted fraction can be found by subtracting reflected fraction from unity. [∵ By energy conservation, transmitted and reflected energy must add up to 1]. Thus, transmitted fraction:

$$\eta_t = 1 - \eta_r = 1 - \left(\frac{v_2 - v_1}{v_2 + v_1}\right)^2$$

$$\Rightarrow \boxed{\eta_t = \frac{4 v_1 v_2}{(v_1 + v_2)^2}}$$

→ It can be easily seen that reflected and transmitted fraction are same for $\frac{v_2}{v_1} = \lambda$ or $\frac{v_1}{v_2} = \lambda$ where λ is some constant. this can be easily seen:

$$\eta_r = \left(\frac{v_2 - v_1}{v_2 + v_1}\right)^2 = \left(\frac{v_2/v_1 - 1}{v_2/v_1 + 1}\right)^2 = \left(\frac{v_1/v_2 - 1}{v_1/v_2 + 1}\right)^2,$$

though if $v_2 < v_1$, there will be a phase change of π in reflected wave while if $v_1 < v_2$ there will be no phase shift.

→ In case of sinusoidal waves, $P = \frac{1}{2} \mu v A^2 \omega^2$ thus:

Incident Power: $P_i = \frac{1}{2} \mu v A_i^2 \omega^2$

Reflected Power: $P_r = \frac{1}{2} \mu v A_r^2 \omega^2$, Thus:

Reflected fraction: $\eta_r = \left(\frac{A_r}{A_i}\right)^2 = \left(\frac{v_2 - v_1}{v_2 + v_1}\right)^2 \Rightarrow \boxed{A_r = \left(\frac{v_2 - v_1}{v_2 + v_1}\right) A_i}$

Similarly we can show that transmitted amplitude:

$$\boxed{A_t = \frac{2 v_2 A_i}{v_1 + v_2}}$$

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EXAMPLE: An incident wave travels at $4\,ms^{-1}$ hitting a boundary such that 25% of the energy is reflected. Amplitude of incident wave is 2 cm, calculate (a) Possible wave speeds across the boundary (b) Possible amplitudes of transmitted wave (c) Amplitude of reflected wave.

Solution: $\eta_R = \frac{1}{4} = \left(\frac{v_2 - v_1}{v_2 + v_1}\right)^2 \Rightarrow \left(\frac{v_2 - 4}{v_2 + 4}\right)^2 \Rightarrow v_2 = 12\,ms^{-1}$ or $\frac{4}{3}\,ms^{-1}$

If $v_2 = 12\,ms^{-1}$, it is a case of reflection from rarer medium, with $\frac{4}{3}\,ms^{-1}$ it will be case case of reflection from denser medium [π shift).

Now transmitted amplitude:

$A_t = \frac{2v_2}{v_1 + v_2} A_i$, For rarer medium, $A_t = \frac{2 \times 12}{12+4} \times 2 = 3$ cm

For denser medium, $A_t = \frac{2 \times 4/3}{4 + 4/3} \times 2 = 1$ cm

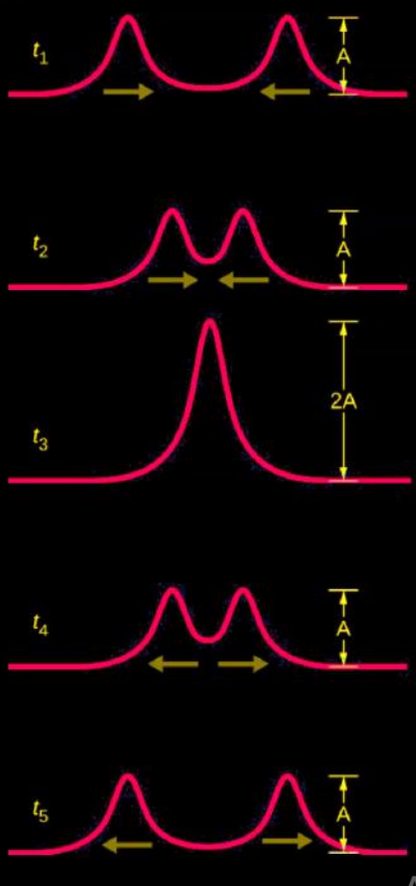
(In one case amplitude is $\frac{1}{2}$ times, in other case 3/2 times).

Now reflected amplitude:

$$A_r = \left|\frac{v_2 - v_1}{v_2 + v_1}\right| A_i = 2 \times \left(\frac{8}{16}\right) = 1\text{ cm}.$$

SUPERPOSITION AND INTERFERENCE:

Consider two pulses moving towards each other as shown in the figure. As the motion progresses, at some point of time, the waves overlap producing a wave with twice the amplitude and continue as if they have passed through each other unaffected by mutual interaction. These pulses are said to have interfered and the phenomenon is known as interference. For mechanical waves principle of superposition states that if two or more travelling waves combine at the same point then resulting motion of the mass element at that location is algebraic sum of the motion



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due to individual waves (by motion here we may mean displacement, velocity acceleration or even slope of the string segment).

Principle of superposition can be understood as follows:

The partial differential equation governing a one dimensional wave is $\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial t^2}$, Recall that if $f(x-vt)$ is a solution to this equation and $g(x+vt)$ is also a also to this equation then

$f(x-vt) + g(x+vt)$ is also a solution to this equation. Each solution corresponds to its own initial conditions, thus $f(x-vt)$ corresponds to initial conditions $y_0 = f_0(x)$ and $\dot{y}_0 = \dot{f}_0(x)$

similarly $g(x+vt)$ corresponds to initial conditions $y_0 = g_0(x)$ and $\dot{y}_0 = \dot{g}_0(x)$. If both waves are simultaneously generated, the net initial conditions become $y_0 = f_0 + g_0$ and $\dot{y}_0 = \dot{f}_0 + \dot{g}_0$ and the superposed solution satisfies this initial condition due to linear nature of partial differential equation (i.e there are no squares or other powers on derivatives).

→ For obeying the superposition principle, mechanical waves must have small amplitudes. If amplitude is too large, medium is distorted past the limit where the restoring forces are linearly proportional to the deformations.

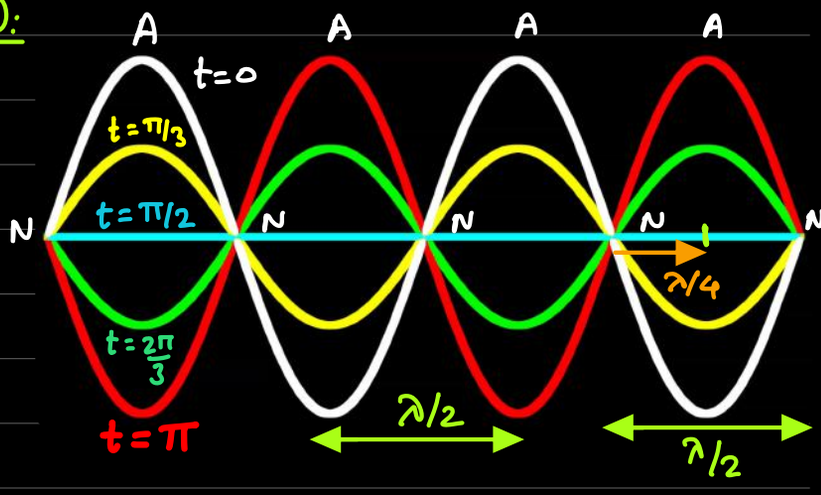
STATIONARY WAVES (STANDING WAVES):

Whenever two waves of same amplitude travelling in opposite directions superpose we get what are known as stationary waves or standing waves. For instance suppose we have two waves given by:

$y_1 = A\sin(kx - \omega t)$ and

$y_2 = A\sin(kx + \omega t)$ then our resultant wave is given as

$y = y_1 + y_2 = 2A\sin kx \cos(\omega t)$.



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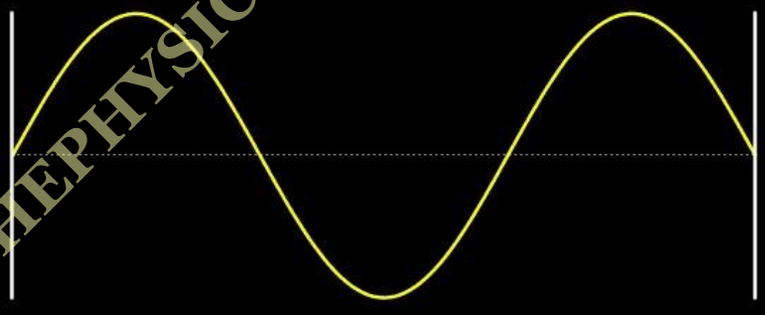
Here following observations can be made:

→ Particles where $\sin kx = 0$ never move from their location and are thus stationary points. Such points are called NODES (N). Thus for location of nodes, $kx = \frac{2\pi}{\lambda} x = n\pi \Rightarrow x = n\lambda/2$. Thus nodes are located at $x = 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}$ ... etc. Node to node separation is $\lambda/2$.

→ Particles for which $\sin kx = \pm 1$ oscillate in the maximum possible amplitude i.e 2A. Such points are called antinodes, for antinodes, $kx = \frac{2\pi}{\lambda} x = (2n+1)\frac{\pi}{2}$ or $x = (2n+1)\frac{\lambda}{4}$. Thus antinodes (A) are located at $x = \lambda/4, 3\lambda/4, 5\lambda/4$ etc... The antinode to antinode separation is also $\lambda/2$. Antinodes are are located at mid points of any two successive nodes thus distance between an antinode and nearest node is $\lambda/4$.

→ Recall that Power transmission in a wave is given as $P = -T \frac{\partial y}{\partial t} \frac{\partial y}{\partial x}$



STATIONARY WAVE ANIMATION

now at antinodes $\partial y/\partial x = 0$ and at nodes $\partial y/\partial t = 0$ thus there is no power transmitted across any node or antinode and net energy remains trapped between any node and successive antinode, hence the name "standing" waves. Energy is standing between nodes and antinodes.

→ When the particles of string are at mean position, maximum energy occurs near antinode (entire energy is in form of KE), when the string is at extreme position, maximum energy occurs at nodes (entire energy is in form of potential energy).

→ As the string moves from extreme to mean energy flow is towards antinode and as it moves from mean to extreme, flow is from antinode to node.

→ The segment of string at a distance of $\lambda/8$ from a node or antinode has constant energy density. Recall when two oppositely travelling waves superpose net energy per unit length is $e_l = e_{l_1} + e_{l_2}$. Now the two waves are $A\sin(kx - \omega t)$ and $A\sin(kx + \omega t)$

Now for first wave:

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$$e_{e_1} = \frac{1}{2}\mu\left[\left(\frac{\partial y}{\partial t}\right)^2 + v^2\left(\frac{\partial y}{\partial x}\right)^2\right] = \mu A^2\omega^2\cos^2(kx-\omega t)$$

Similarly $e_{e_2} = \mu A^2\omega^2\cos^2(kx+\omega t)$

Now $e_e = e_{e_1} + e_{e_2} = \frac{\mu^2 A^2\omega^2}{2}\left[\cos 2(kx-\omega t) + 1 + \cos 2(kx+\omega t) + 1\right]$

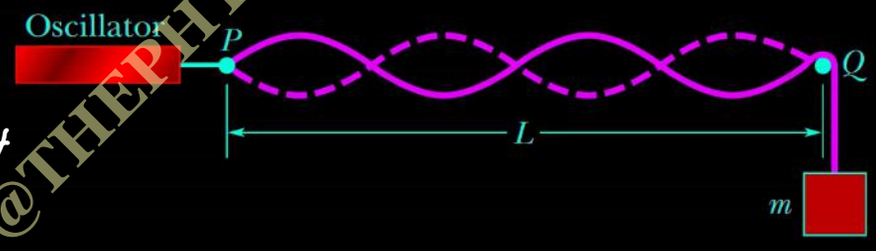
Now @ $x = \lambda/8$, $kx = \frac{2\pi}{\lambda}\cdot\frac{\lambda}{8} = \pi/4$, thus:

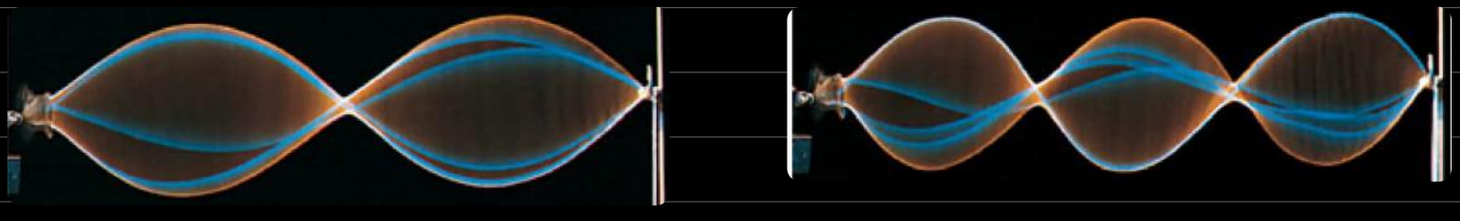
$$e_e = \frac{\mu A^2\omega^2}{2}\left[2 + \cos\left(\frac{\pi}{2} - 2\omega t\right) + \cos\left(\frac{\pi}{2} + 2\omega t\right)\right] = \mu A^2\omega^2$$

thus e_e at $\lambda/8$ is independent of time. Similarly energy density at all points at bisectors of a node and consecutive antinode are independent of time.

RESONANCE (STANDING WAVES IN STRINGS):

Imagine your friend swinging on a swing in a garden with a time period T then if you want the swing to swing in large amplitudes, then you can't push the swing at arbitrary intervals, you should be pushing it at regular intervals that are multiples of T. Similarly suppose we want a string to have stationary wave of large amplitude, the oscillator has to provide the energy to the string at certain frequencies. These frequencies are called natural frequencies or resonant frequencies of the system. At these frequencies, the systems oscillate with well defined nodes and antinodes, however if energy is provided to the system at any other frequency, the system undergoes erratic oscillations.





STROBOSCOPIC PHOTOGRAPHS AT NEAR RESONANCE FREQUENCIES (SECOND HARMONIC AND THIRD HARMONIC).

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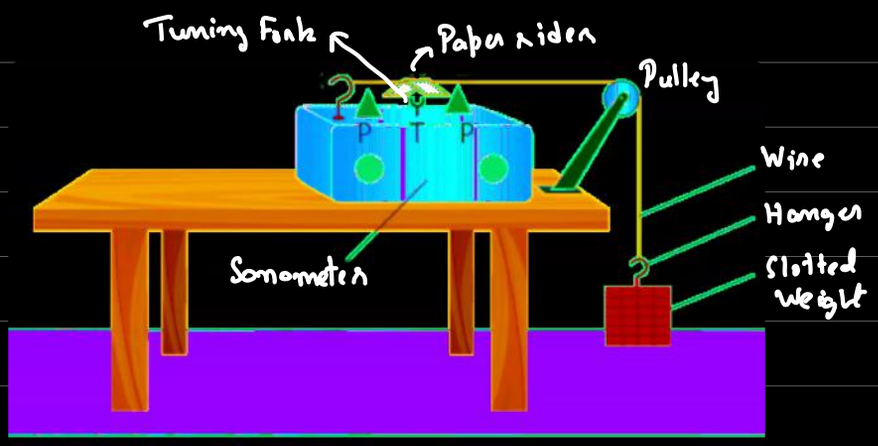
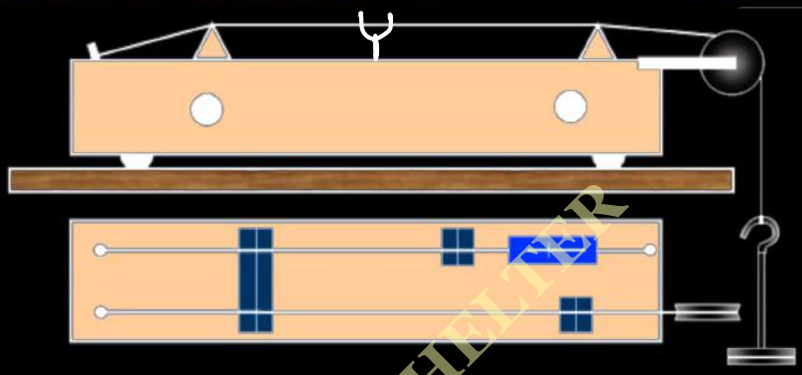
SONOMETER: It is a device for studying standing waves in taut strings. It has arrangement for varying tension in the string and some arrangement for setting up forced oscillations in the string. Usually three kinds of mechanisms are common:





1. Tuning Fork
2. AC mains while passing current through sonometer wire (Forcing using magnetic force).
3. Electromagnet driven by AC mains (Forcing a ferromagnetic wire).

→ While forcing a ferromagnetic wire using an electromagnet, the forcing frequency is double that of AC Mains.

→ To figure out the length at which resonance occurs, a paper rider is put on the vibrating wire and effective vibrating length is adjusted using the bridges. (Numerical examples will follow later on for better understanding).

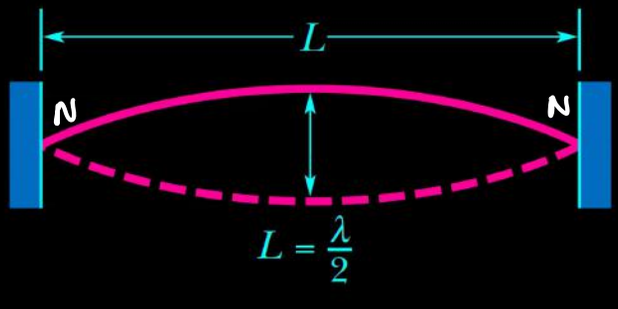
RESONANT FREQUENCIES OF STRING FIXED AT BOTH ENDS:

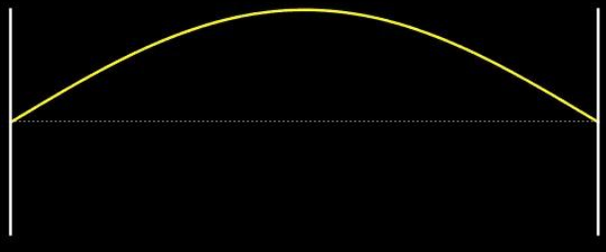
FIRST HARMONIC (FUNDAMENTAL):

Consider neat oscillations of the string in a single loop (like a guitar string).

Here $L = \frac{\lambda}{2} = v/2\nu$

$$\Rightarrow \boxed{\nu = \frac{v}{2L}}$$ ← Fundamental Frequency.





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The least frequency for which resonance occurs is called fundamental and all its multiples are called harmonics. Thus if fundamental frequency is 50 Hz then 100 Hz will be called second harmonic, 150 Hz will be called 3rd harmonic and so on.

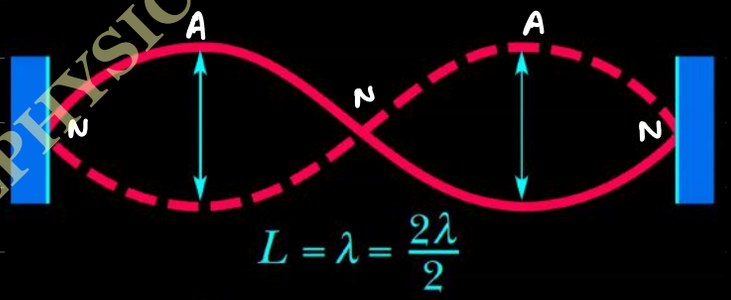
Overtone Number: If all natural frequencies of a system are arranged in ascending order and numbered sequentially beginning with zero, then number corresponding to each frequency is called overtone number. For example if natural frequencies of a system are 50 Hz, 100 Hz, 150 Hz etc then 50 Hz is zeroth overtone (or fundamental), 100 Hz is first overtone, 150 Hz second overtone number and so on.

Second Harmonic (First Overtone):

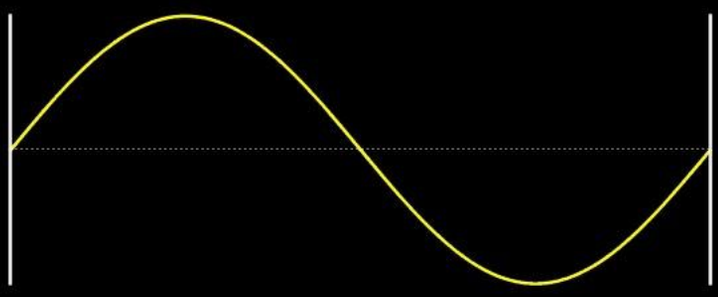
Here the string oscillates in two neat loops as shown. Now:

$$L = 2\lambda/2 = 2v/2\nu$$

$$\Rightarrow \quad \nu = \frac{2v}{2L}$$



$$L = \lambda = \frac{2\lambda}{2}$$



this just double the frequency of the fundamental.

(Recall that energy per unit length is simply the sum of energy of component waves).

Third Harmonic And Higher Harmonics:

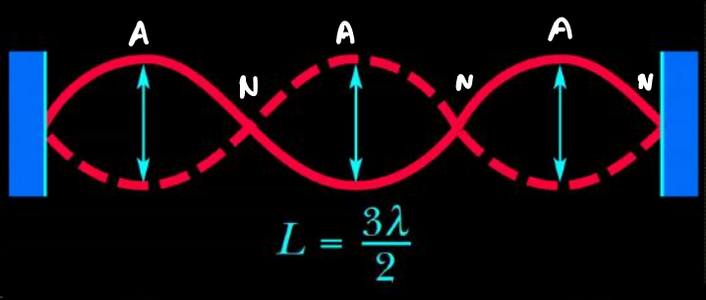
In the third harmonic (Second Overtone), the string oscillates in three neat loops. Thus: $L = \frac{3\lambda}{2} = \frac{3v}{2\nu}$

$\Rightarrow \quad \nu = 3v/2L$, continuing the pattern, for n^{th} harmonic:

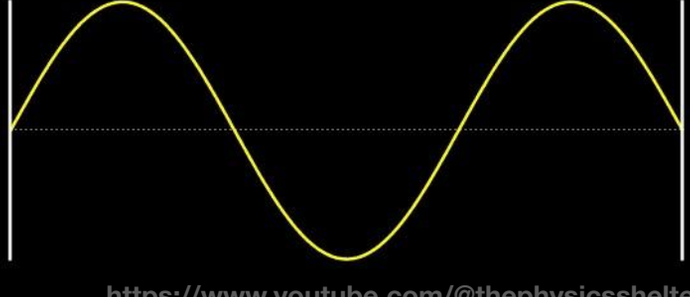
$$\boxed{\nu = \frac{nv}{2L}}$$

Harmonic Number: n

Overtone Number: $n-1$



$$L = \frac{3\lambda}{2}$$



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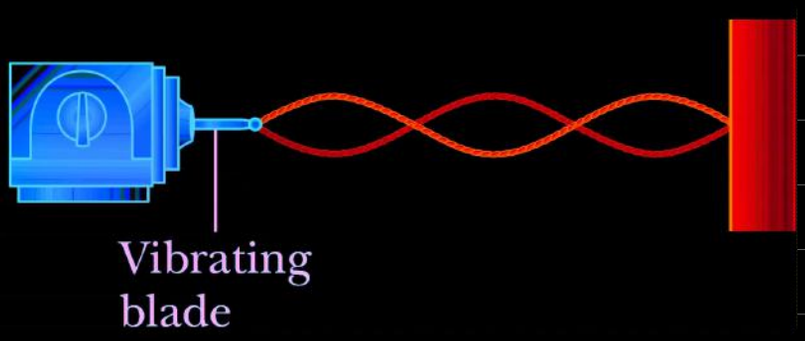
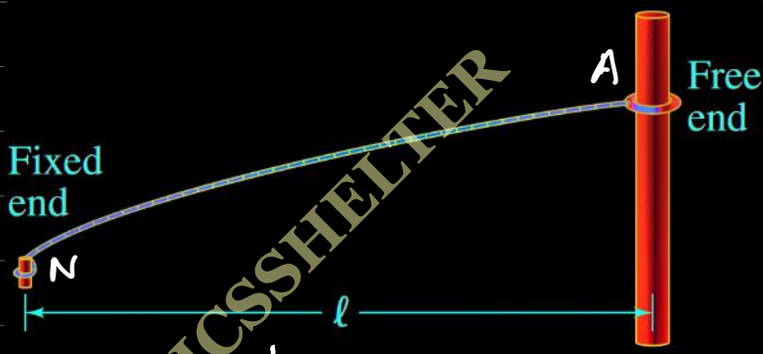
Instead of sonometer some labs also use an electrical oscillator driven by a motor whose rpm can be controlled to get desired excitation frequency.



Resonance Frequencies For String Fixed At One End And Free At Other:

Stationary waves can be created in strings with one free end by attaching an oscillator at the other end which creates oscillation of slope. For instance adjacent figure shows fundamental mode of such an oscillation.

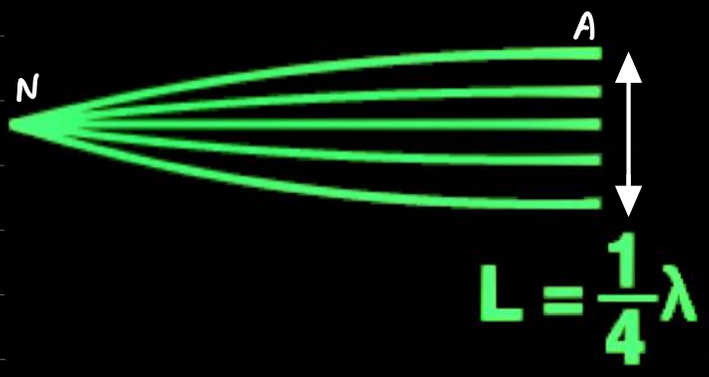


Fundamental Frequency Of String With One End Free:

The adjacent figure shows the oscillation in fundamental mode (Half loop).

Here, $L = \lambda/4 = v/4\nu$

$\Rightarrow$ $\boxed{\nu = v/4L}$ ← Fundamental



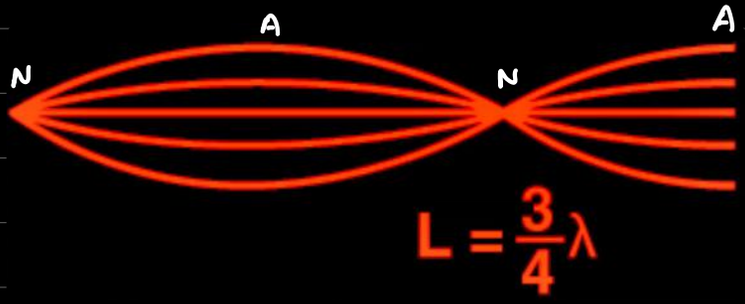
First Overtone (Third Harmonic) (String free at one end):

The first overtone configuration has 1.5 loops as shown in the figure,

here, $L = 3\lambda/4 = 3v/4\nu$

$\Rightarrow$ $\nu = 3v/4L$

Note that here this is an odd multiple of fundamental, the even multiple (2) is not a natural frequency.



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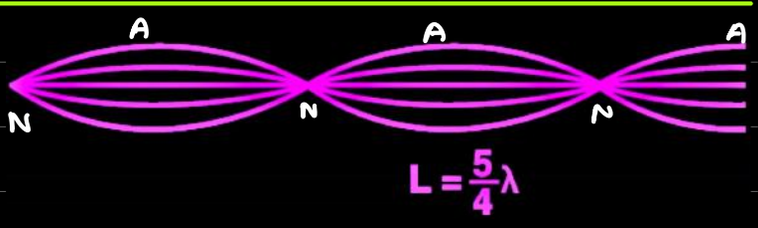
SECOND OVERTONE AND GENERAL RESONANCE FREQUENCIES (ONE END FIXED):



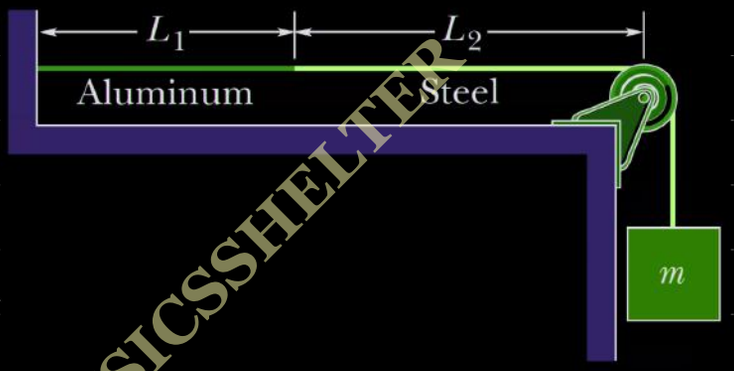
The second overtone oscillation is illustrated in adjacent figure, here, $L = 5\lambda/4 = 5v/4\nu$

$\Rightarrow \nu = 5v/4L$, thus continuing this pattern we will get:

$$\boxed{\nu = (2n+1)\frac{v}{4L}}$$

n: OVERTONE NUMBER

$2n+1$: HARMONIC NUMBER.



EXAMPLE: In the shown figure, an Aluminum wire of length $L_1 = 60$ cm, cross sectional area 1×10^{-2} cm^2 and density 2.6 g/cc is joined to steel wire of density 7.8 g/cc and same cross sectional area. The combined wire is loaded with $m = 10$ kg so that distance L_2 from joint to pulley is 86.6 cm. Transverse waves are setup on the wire by an external source of variable frequency (with node at joint).

(a) Find lowest frequency such that standing waves are formed with joint as a node.

(b) How many total nodes are observed at this frequency?

Solution: Since there are nodes at both ends of Aluminum portion as well as steel portion, they behave like strings fixed at both ends accordingly the required resonance frequency must satisfy the resonant frequency condition for both wires. Let ν is the required resonant frequency then:

$$\nu = \frac{n_1 v_A}{2L_1} = \frac{n_2 v_s}{2L_2} \quad (1)$$ where n_1 and n_2 are natural numbers.

and v_A and v_s are wave velocities in aluminum section and steel section respectively. Also $v_A \propto \sqrt{\frac{T}{\mu_A}} = \sqrt{\frac{T}{\rho_A S}}$ i.e $\propto \frac{1}{\sqrt{\rho_A}}$, similarly $v_s \propto \frac{1}{\sqrt{\rho_s}}$.

Now from (1), $\frac{n_1}{n_2} = \frac{v_s}{v_A}\cdot\frac{L_1}{L_2} = \frac{L_1\sqrt{\rho_A}}{L_2\sqrt{\rho_s}} = \frac{60}{86.6}\sqrt{\frac{2.6}{7.8}} = \frac{2}{5}$

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thus lowest values satisfying this condition are $n_1 = 2$, $n_2 = 5$. Accordingly, Aluminum string oscillates in two loops and steel strings oscillates in 5 loops. Now required frequency:

$$\gamma = \frac{2V_A}{2L_1} = \sqrt{\frac{mg}{\rho_A \times S}} \cdot \frac{1}{L_1}$$

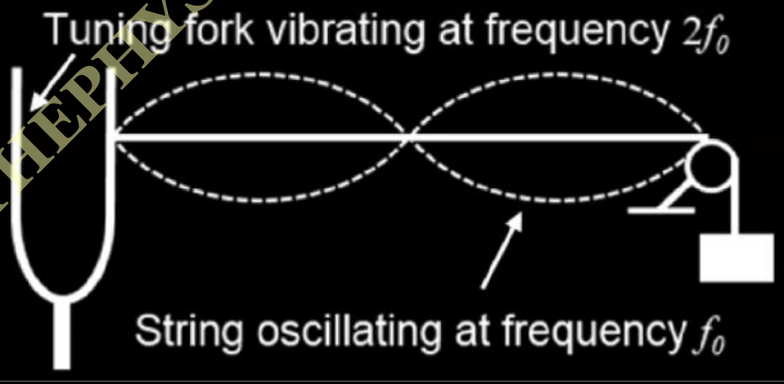
$$\Rightarrow \gamma = \frac{1}{0.6}\sqrt{\frac{10 \times 9.8}{2.6 \times 10^3 \times 10^{-6}}} \quad \text{(Remember to convert to SI units)}$$

$$\Rightarrow \gamma = 324\ \text{Hz} \quad \text{Ans.}$$

Now we have a total of $2+5 = 7$ loops, thus number of nodes $= 7+1 = 8$ nodes (Counting the end points).

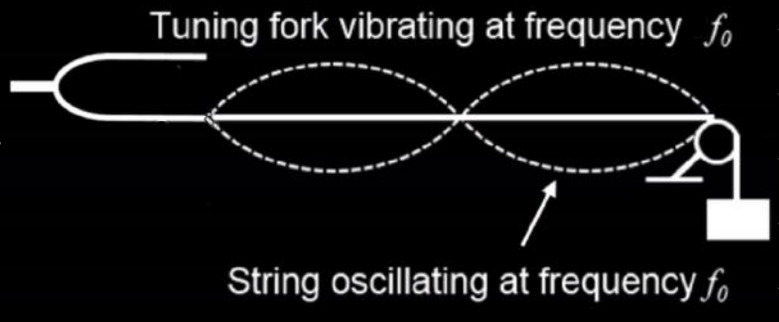
MELDE'S EXPERIMENT:

Melde's experiment is another setup for studying stationary waves in strings. In modern setup it consists of an electrically driven tuning fork and the setup can be made in two ways:



VERTICAL PRONGS MODE.

① Vertical prongs: In this mode the prongs of tuning fork oscillate left and right while string oscillates up and down at half the frequency of the tuning fork. When the prong is at left extreme, the string is at mean position and when the prong is at right, the wire is at either upper extreme or lower extreme.



HORIZONTAL PRONGS MODE

② Horizontal prongs: In this setup, the prongs oscillate up and down and the frequency of oscillation is same for the string and the tuning fork.

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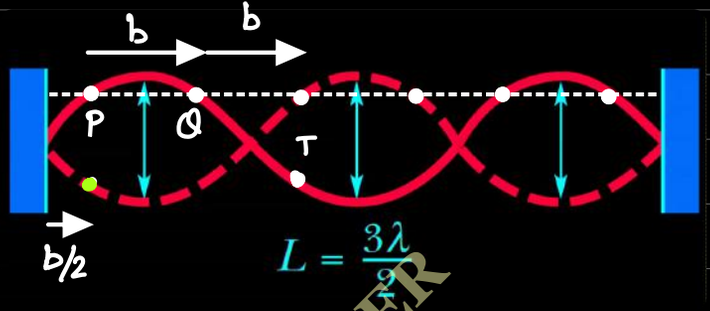
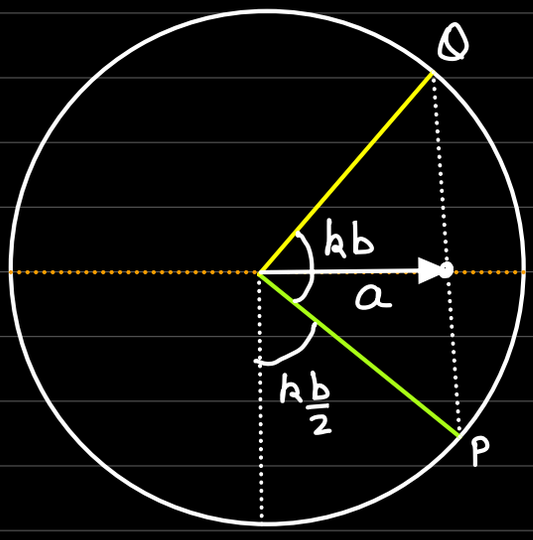
EXAMPLE: IRODOV 4.166

A string 120 cm in length sustains a standing wave, with the points of the string at which the displacement amplitude is equal to 3.5 mm being separated by 15.0 cm. Find the maximum displacement amplitude. To which overtone do these oscillations correspond?

Solution: We are looking at some special points where amplitude is same and the particles are equidistant from each other. (see figure), also since maximum amplitude is asked, these points can't themselves be points of maximum amplitudes (i.e these can't be antinodes). Now let amplitude vary as $2A\sin kx$. The amplitude can be visualised as a phasor revolving at the rate k radians per metre (Just like usual phasors rotate at ω rad/s' here we are visualising a phasor rotating with space instead of time). Now we want the rotation angle of phasor in distance b such that projection after travelling a distance b is same.



Refer to the phasor diagram (drawn with space angle kx instead of time angle 'ωt').



According to phasor diagram:

$$2A\sin\left(\frac{kb}{2}\right) = 2A\cos(kb/2)$$

$$\Rightarrow \frac{kb}{2} = \pi/4 \Rightarrow b = \frac{\pi}{2k} = \frac{\pi}{2\times 2\pi/\lambda}$$

$\Rightarrow b = \lambda/4$. Thus these points are at a distance of $\lambda/4$ from each other and at exact bisector of node and antinodes (See figure, first such point is at $b/2$ from one first node i.e $\lambda/8$).

For our case, given $b = 15\text{ cm} = \lambda/4 \Rightarrow \lambda = 60$ cm. Now since each loop has a length of $\lambda/2$ i.e 30 cm thus in a string of length 120 cm, 4 such loops will be formed. Thus these oscillations correspond to 4th harmonic or the third overtone).

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Now $2A\sin kx$ at $x = b/2$ is given as 3·5mm (amplitude at these points) Thus:

$$2A\sin\left(\frac{\pi}{4}\right) = 3{\cdot}5\text{mm} \Rightarrow 2A = 3{\cdot}5\times\sqrt{2}\text{ mm} = 4{\cdot}94\text{mm}$$

$\Rightarrow$ Thus required amplitude = 5mm.

EXAMPLE: IRODOV 4·160: In a progressive wave at some point energy density is u_0 at a time T/6 after the particle passes the displacement maxima. Find the average energy density $\langle u\rangle$ in the string.

Solution: Let the wave be $A\cos(\omega t - kx)$ and let us consider point located at $x=0$. This point has maximum displacement at $t=0$ itself. Now energy density:

$$e_k = \frac{1}{2}\mu\left[\left(\frac{\partial y}{\partial t}\right)^2 + v^2\left(\frac{\partial y}{\partial x}\right)^2\right] = \mu\left(\frac{\partial y}{\partial t}\right)^2 \text{ (for progressive).}$$

For energy per unit volume, $u_k = \rho\left(\frac{\partial y}{\partial t}\right)^2$ $(\because \mu/s = \rho)$.

Now $\partial y/\partial t = -A\omega\sin(\omega t - kx) \Rightarrow u_k$ @ $t = T/6$, $x = 0$:

$$u_k = \rho A^2\omega^2\sin^2\left(\frac{\omega T}{6}\right) = \rho A^2\omega^2\sin^2\left(\frac{2\pi}{6}\right) = \frac{3}{4}\rho A^2\omega^2 = u_0$$

Also $\langle e_k\rangle = \frac{1}{2}\mu A^2\omega^2 \Rightarrow \langle u_k\rangle = \frac{1}{2}\rho A^2\omega^2 = \frac{1}{2}\times\frac{4}{3}u_0$

$$\Rightarrow \langle u\rangle = \frac{2}{3}u_0 \text{ Ans.}$$

EXAMPLE: IRODOV 4·167:

Find the ratio of the fundamental tone frequencies of two identical strings after one of them was stretched by $\eta_1 = 2.0\%$ and the other, by $\eta_2 = 4.0\%$. The tension is assumed to be proportional to the elongation.

Solution: $\nu = \frac{v}{2\ell} = \sqrt{\frac{T}{\mu}}\cdot\frac{1}{2\ell} = \sqrt{\frac{T\ell}{m}}\cdot\frac{1}{2\ell} = \sqrt{\frac{T}{m\ell}}\cdot\frac{1}{2}$

$$\Rightarrow \frac{\nu_1}{\nu_2} = \sqrt{\frac{T_1}{T_2}\frac{\ell_2}{\ell_1}} = \sqrt{\frac{\eta_1}{\eta_2}\times\frac{\ell_0(1+\eta_2)}{\ell_0(1+\eta_1)}} \quad (\text{where } \ell_0 \text{ is unstretched length})$$

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$$\Rightarrow \quad \nu_1/\nu_2 = \sqrt{\eta_1(1+\eta_2)/\eta_2(1+\eta_1)} \quad \Rightarrow \quad \frac{\nu_2}{\nu_1} = \sqrt{\frac{\eta_2(1+\eta_1)}{\eta_1(1+\eta_2)}} = 1.4 \text{ Ans.}$$

EXAMPLE: IRODOV 4.168

Determine in what way and how many times will the fundamental tone frequency of a stretched wire change if its length is shortened by 35% and the tension increased by 70%.

Solution: $\nu = \frac{v}{2l} = \sqrt{\frac{T}{\mu}} \cdot \frac{1}{2l} \Rightarrow \frac{\nu_2}{\nu_1} = \sqrt{\frac{T_2}{T_1}} \cdot \left(\frac{l_1}{l_2}\right)$

Now $\frac{l_1 - l_2}{l_1} = 0.35 \Rightarrow 0.65 l_1 = l_2 \Rightarrow \frac{l_1}{l_2} = \frac{1}{0.65}$

Thus $\frac{\nu_2}{\nu_1} = \sqrt{\frac{1.7}{1}} \times \frac{1}{0.65} = 2.006 \approx 2$ Ans.

EXAMPLE : IRODOV : 1.172:

A string of mass m is fixed at both ends. The fundamental tone oscillations are excited with circular frequency ω and maximum displacement amplitude a_{max}. Find :

(a) the maximum kinetic energy of the string;

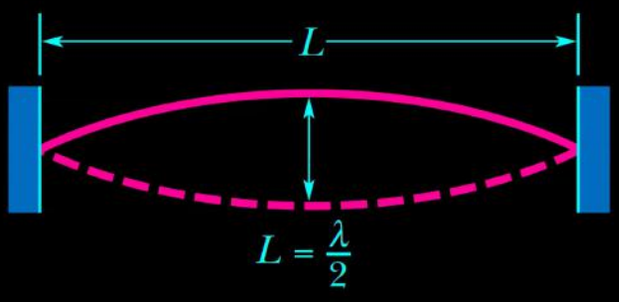
(b) the mean kinetic energy of the string averaged over one oscillation period.

Solution: Let wave be $a_{max} \sin kx \cos \omega t = y$

then :

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{2L} = \pi/L$$

Thus $y = a_{max} \sin\left(\frac{\pi x}{L}\right) \cos(\omega t)$



Now consider an element of length dx at a distance x, dKE:

$$dKE = \frac{1}{2}\left(\frac{m}{L}dx\right)\left(\frac{\partial y}{\partial t}\right)^2 = \frac{1}{2}\frac{m}{L}dx \cdot a_{max}^2 \omega^2 \sin^2(\omega t) \sin^2\left(\frac{\pi x}{L}\right) \quad (1)$$

This is max when $\sin \omega t = \pm 1$, thus $KE_{max} = \frac{m}{2L} a_{max}^2 \omega^2 \int_0^L \sin^2\left(\frac{\pi x}{L}\right) dx$

$$\Rightarrow KE_{max} = \frac{1}{4} m a_{max}^2 \omega^2 \quad (2) \text{ Ans.}$$

Now from (1) & (2), we can write instantaneous KE as:

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$KE = \frac{1}{4} m a_{max}^2 \omega^2 \times \sin^2(\omega t)$, thus $\langle KE \rangle = \frac{1}{4} m a_{max}^2 \omega^2 \langle \sin^2 \omega t \rangle$

$\Rightarrow \langle KE \rangle = \frac{1}{8} m a_{max}^2 \omega^2$ Ans.

EXAMPLE: YOUNG & FREEDMAN:

15.39 • CALC A thin, taut string tied at both ends and oscillating in its third harmonic has its shape described by the equation $y(x, t) = (5.60 \text{ cm}) \sin[(0.0340 \text{ rad/cm})x] \sin[(50.0 \text{ rad/s})t]$, where the origin is at the left end of the string, the x-axis is along the string, and the y-axis is perpendicular to the string. (a) Draw a sketch that shows the standing-wave pattern. (b) Find the amplitude of the two traveling waves that make up this standing wave. (c) What is the length of the string? (d) Find the wavelength, frequency, period, and speed of the traveling waves. (e) Find the maximum transverse speed of a point on the string. (f) What would be the equation $y(x, t)$ for this string if it were vibrating in its eighth harmonic?

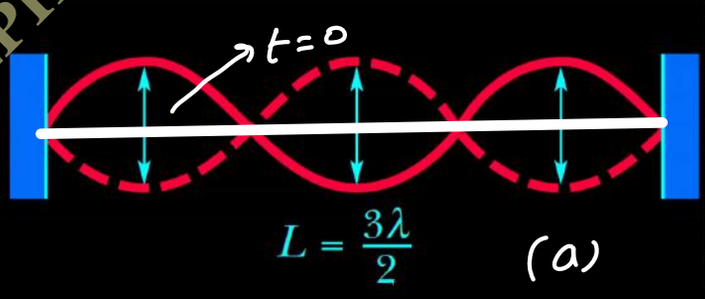
Solution: Wave is of the form:

$y = 2A \sin kx \sin(\omega t)$, converting from product to sum form,

$y = A\cos(kx - \omega t) - A\cos(kx + \omega t)$

thus component waves are:

$y_1 = A\cos(kx - \omega t)$ and $y_2 = -A\cos(kx + \omega t)$



By comparison, $2A = 5.6$ cm $\Rightarrow A = 2.8$ cm Ans (Amplitude of components) (b)

By comparison $k = 0.0340$ rad/cm $= \frac{2\pi}{\lambda} \Rightarrow \lambda = 184.8$ cm. (d)

Now length of string: $L = 3\lambda/2 \approx 277$ cm. Ans. (c)

By comparison, $\omega = 50 \text{ rad s}^{-1} \Rightarrow \nu = \frac{\omega}{2\pi} = 7.95 \text{ Hz}$, $T = \frac{2\pi}{\omega} = 0.126$ sec (d)

$v = \omega/k = 50/0.0340 \text{ cm s}^{-1} = 14.7 \text{ m s}^{-1}$ (d) Ans.

Now max transverse speed: $2A\omega = 5.6 \times 50 \text{ cm s}^{-1} = 2.8 \text{ m s}^{-1}$ (e) Ans.

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(f) Right now string is oscillating in 3rd harmonic for oscillating in 8th harmonic, wavelength will become 3/8 times and ω will become 8/3 times ($\because \nu = nv/2l$). Thus wave equation for 8th Harmonic:

$$y = 5.60\ \text{cm}\ \sin\left(\frac{8}{3} \times 0.0340\ \text{rad/cm}\right) \sin\left(50 \times \frac{8}{3}\ \text{rad s}^{-1} t\right)$$

Thus $y = 5.60\ \text{cm}\ \sin(0.0907\ \text{rad/cm})\ \sin(133.33\ \text{rad s}^{-1} t)$ (f) ans.

($\because \lambda$ becomes 3/8 times, k becomes 8/3 times).

EXAMPLE: A light string is tied at one end to a fixed support and to a heavy string of equal length L at the other end as shown in the figure. Mass per unit length of strings are μ and 9μ respectively with tension of composite string equal to T. Find the possible frequencies for which standing waves are formed with joint being a node.

Solution: $\nu_1 = \frac{n_1 v_1}{2L} = n_1 \sqrt{\frac{T}{\mu}} \cdot \frac{1}{2L}$ similarly $\nu_2 = n_2 \sqrt{\frac{T}{9\mu}} \times \frac{1}{2L}$

$$\Rightarrow \nu_2 = \frac{n_2}{3} \sqrt{\frac{T}{\mu}} \cdot \frac{1}{2L}$$

Now Since both strings are oscillating with same frequency, thus:

$\nu_1 = \nu_2 \Rightarrow n_1 = n_2/3$, thus n_2 must be a multiple of 3.

thus required set of resonance frequencies:

$$\nu = \frac{n}{2L} \sqrt{\frac{T}{\mu}} \quad \text{ans.}$$

→ Note for LCM of fractions we do: LCM of Numerators / HCF of Denominators,

→ In general for this type of questions we need LCM of fundamental frequencies of two parts.

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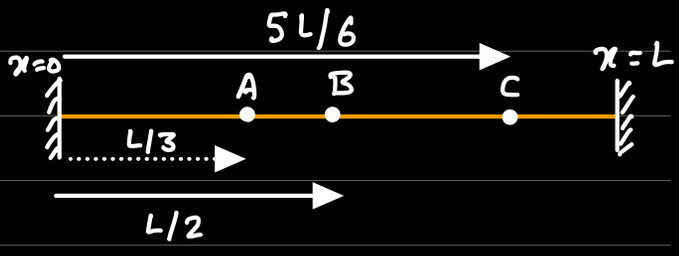
Example:

Standing waves are established on a string of length L such that A is a node and B is immediate antinode. Oscillation amplitude for point B is a_0. Find the amplitude as a function of coordinate x as measured from left end.



Solution: Node to antinode distance $AB = \lambda/4 = L/2 - L/3 = L/6$

$\Rightarrow \lambda = 2L/3$.

Now if the wave be $a_0 \sin kx \cos\omega t$, $k = \frac{2\pi}{\lambda} = \frac{2\pi}{2L} \times 3 = \frac{3\pi}{L}$

Thus amplitude as a function of x:

$$A = a_0 \sin\left(\frac{3\pi x}{L}\right) \text{ Ans.}$$

Another Way Of Looking At Stationary Waves:

Consider a pulse generated on a string travelling on a string with a speed v. The round trip time for this pulse is $2L/v$. At this instant if we provide a reinforcement, the pulse will get a larger amplitude. Thus fundamental time period: $T_0 = 2L/v$

Therefore fundamental frequency: $\nu_0 = \frac{1}{T_0} = v/2L$.

Now if we provide reinforcements at any whole number fraction of fundamental time period, it will create extra pulse in between however when fundamental time period is completed, the first pulse will again get reinforced, thus any fraction of fundamental time period also leads to large amplitude waves.

Thus $T_n = \frac{2L}{v} \times \frac{1}{n} \Rightarrow \boxed{\nu_n = nv/2L}$ this is the same result that we derived earlier.

Example: A certain string has mass density per unit length varying as $\mu = \mu_0\left(1 + \frac{x}{L}\right)^2$ where x is the coordinate measured from left end.

Find set of its natural frequencies. (Tension is T, length is L)

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Solution: $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{\mu_0}} \frac{1}{(1+x/L)}$, now round trip time:

$$T_0 = \int \frac{dx}{v} = \left[\sqrt{\frac{\mu_0}{T}} \int_0^L (1+x/L)\,dx\right] \times 2 = 3L\sqrt{\frac{\mu_0}{T}}$$

$\Rightarrow$ Resonant time periods: T_0/n $\Rightarrow$ resonant frequencies $= n/T_0$

Thus resonance frequencies:

$$\nu_n = \frac{3n}{L}\sqrt{\frac{T}{\mu_0}} \quad \text{Ans.}$$

EXAMPLE: Find set of natural frequencies of a string fixed at both ends such that a point at $L/4$ from from one of its ends is an antinode. The length of the string is L

Solution: For an antinode, distance from left end must be $(2n+1)\lambda/4$

thus: $\frac{L}{4} = (2n_1+1)\lambda/4$ (1) also string must have integer number of half loops thus, $L = n_2\lambda/2$ (2),

From (1) & (2):

$$4 = \frac{2n_2}{(2n_1+1)} \Rightarrow n_2 = 2(2n_1+1)$$

thus $\lambda = \frac{2L}{n_2} = \frac{2L}{2(2n_1+1)} = \frac{L}{2n_1+1} = \frac{v}{\nu}$

$\Rightarrow \nu = \frac{v}{L}(2n+1)$

THINK YOURSELF: Suppose minimum frequency with this condition were 100 Hz, what would be the next higher frequency?

Ans: $\frac{v}{L} = 100\ Hz \Rightarrow \frac{3v}{L} = 300\ Hz$ Ans.

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